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RESEARCH ARTICLE

Research on Multi-AGV Rail Transportation Path Planning and Dynamic Obstacle Avoidance Based on Multi-Sensor Environmental Perception and Deep Reinforcement Learning

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ABSTRACT Sensor-equipped Automated Guided Vehicles (AGVs) are widely used in material distribution and other logistics processes. Path planning, a critical component of rail transport material handling systems, ensures the efficiency and safety of AGV material transport. Traditional fixed-path AGV transport is costly to maintain and prone to collisions due to path conflicts in large-scale operations. Addressing these challenges, this study first considers the characteristics of real-world rail transport material handling environments. It utilizes the A* algorithm as the foundation to develop a path planning method for multiple AGVs, enabling global path planning for each AGV. Further, it employs advanced AGVs equipped with multiple sensors to collect and process sensor data, developing a multi-sensor feature fusion environmental perception algorithm. This algorithm effectively identifies other dynamic AGVs, static obstacles, and targets within rail transport environments. Additionally, this research constructs a deep reinforcement learning model based on Deep Q-Networks, designs a novel reward function, and, integrating environmental perception capabilities, achieves dynamic obstacle avoidance for complex multi-AGV path planning. Finally, a simulation environment was established to verify the applicability of the proposed path planning and dynamic obstacle avoidance methods for multiple AGVs in rail transportation scenarios, providing experimental support for railway logistics handling requirements. Benchmark tests demonstrated that the proposed GG-DRL method significantly outperforms the traditional Artificial Potential Field (APF) algorithm, with a 70% reduction in collisions, a 25%–30% decrease in planning time, a 10%–15% reduction in path length, and a 20%–25% decrease in root mean square (RMS) curvature. Further comparative experiments conducted using a standardized Maze Benchmark Map and dynamic obstacle scenarios indicated that the GG-DRL algorithm is markedly superior to both traditional path planning methods such as Rapidly-exploring Random Tree (RRT) and classical deep reinforcement learning approaches such as Deep Deterministic Policy Gradient (DDPG), in terms of path quality, computational efficiency, obstacle avoidance performance, and trajectory smoothness. Specifically, GG-DRL achieved the shortest average path length of 81.12 units, the fastest average computation time of 11.94 seconds, the fewest obstacle avoidance maneuvers with an average of 5.2, and the highest average trajectory smoothness of 0.86. In increasingly complex environments containing 10, 20, and 30 dynamic obstacles respectively, the GG-DRL algorithm consistently exhibited outstanding robustness, planning speed, and trajectory quality. These results confirm the effectiveness of multi-sensor fusion and deep reinforcement learning in enhancing the reliability, efficiency, and safety of AGV transportation, indicating broad application prospects in railway logistics and intelligent navigation.

INDEX TERMS Rail transit material handling, automatic guided vehicles, multi-AGV path planning, heuristic algorithms, deep reinforcement learning, dynamic obstacle avoidance.

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I. INTRODUCTION

In recent years, with the advancement of technological innovation and automation technology, the transportation industry is shifting towards intelligence, informatization, and automation. The integration and development of rail transportation with artificial intelligence are becoming increasingly close. Rail transportation systems, with their advantages of high capacity, comfort, and eco-friendliness, are seeing a notably accelerated pace of construction [1]. To promote the application and development of intelligent and automation technologies in the rail transportation field, the use of Automated Guided Vehicles (AGVs) as tools for material transportation has become a hot topic in the industry. Meanwhile, AGVs are also seen as an indispensable part of the push towards intelligent development in the logistics industry, attracting widespread attention from researchers within the sector.

Against the backdrop of rapid developments in smart logistics and automated warehouse systems, the path planning for multiple Automated Guided Vehicle (AGV) systems has become a critical technological challenge for enhancing system efficiency and safety. Effective path planning must ensure collision-free navigation of AGVs in complex environments, while also seeking to improve planning efficiency and reduce the overall system's energy consumption. To this end, scholars have proposed a series of studies based on traditional path planning algorithms to address the path planning issues in multi-AGV systems. Yang et al. explored a traffic control algorithm for multi-AGV path planning based on system-level scheduling, providing theoretical support and practical verification for the scheduling of multiple AGVs in dock transportation tasks [2]. Pratama et al. aimed to design an efficient multi-agent AGV path planning system based on an improved A* algorithm, effectively managing the path allocation among multiple AGVs to avoid collisions through centralized processing in a server [3]. Hu et al. further optimized the traditional A* algorithm by adding turn weight in the heuristic function and employing a task segmentation strategy, proposing an improved A* algorithm to enhance search efficiency and thoroughness in real working environments [4]. Tien & Nguyen introduced a dynamic path planning method based on updating the weight graph, effectively addressing the operational challenges of multi-AGV systems in medical environments [5]. Furthermore, Huang et al. enhanced the performance of the algorithm in obstacle avoidance and path smoothness by improving the genetic algorithm, effectively increasing the path planning efficiency of the original algorithm [6]. In recent years, the application research of multi-sensor fusion in the field of intelligent vehicle environmental perception has gained widespread attention. Scholars have proposed various fusion strategies and frameworks to improve the accuracy and robustness of the perception system. Shi and others discussed methods to improve multi-sensor fusion effects through spatiotemporal evidence generation and independent visual channels, significantly enhancing the detection and classification accuracy of

vehicle environmental perception. This research highlighted the potential and effectiveness of multi-sensor fusion in real traffic environments [7]. Senel and others proposed a modular, real-time multi-sensor fusion framework specifically for real-time multi-object tracking in intelligent transportation applications, showcasing the framework's ability to continue operating even when individual sensors fail, highlighting the system's adaptability and robustness [8]. Bouain focused on an embedded multi-sensor data fusion design for intelligent vehicle environmental perception tasks, based on a modular and scalable architecture, demonstrating the application potential of sensor fusion technology in resource-constrained environments [9]. Xiang and others reviewed multi-sensor fusion and collaborative perception research in the field of autonomous driving, proposing a new classification method for multi-sensor fusion. This research provides a new perspective for understanding and categorizing different fusion strategies, guiding future research directions [10]. Jayaratne explored a scalable fusion technique based on unsupervised machine learning to improve the environmental perception capabilities of autonomous robots in unknown environments [11]. With the rapid development of smart manufacturing and logistics automation technologies, the path planning problem for multiple Automated Guided Vehicles (AGV) has become one of the research hotspots. Deep reinforcement learning, as a machine learning method capable of acquiring optimal strategies through learning, has shown tremendous application potential and research value in the field of multi-AGV path planning. Liao and others explored a reinforcement learning-based AGV path planning model that effectively guides AGVs to plan optimal paths in complex environments through the Sarsa algorithm combined with a simulated annealing strategy. The proposed potential field method combined with the Deep Q-Network algorithm effectively addresses the shortcomings of traditional reinforcement learning algorithms in dealing with large-scale state spaces, significantly improving the success rate and efficiency of AGV path planning in warehousing logistics [12]. Yin and others proposed a curiosity-driven deep reinforcement learning method to solve the path planning problem of AGVs in real and complex environments. This method uses an enhanced Proximal Policy Optimization (PPO) method and provides additional intrinsic rewards for AGV agents in sparse reward scenarios through a curiosity mechanism, showing that AGVs can efficiently plan reasonable and safe paths [13]. Guo and others introduced an autonomous path planning method for AGVs based on deep reinforcement learning, using a Dueling Double Deep Q-Network with Prioritized Experience Replay (Dueling DDQN-PER) to learn AGV control in a simulated Intelligent Logistics System (ILS) environment, significantly improving the AGV's target approach capability and obstacle avoidance ability [14]. Ye and others developed a multi-agent reinforcement learning algorithm for multiple AGVs aimed at minimizing overall energy consumption. This algorithm,

based on the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm, effectively addresses energy efficiency routing problems providing an effective solution for energy-efficient routing in complex environments [15]. Zhang and others targeted the dynamic path planning problem of AGVs in mixed-flow workshops. Based on reinforcement learning theory, they proposed an improved reinforcement learning algorithm. By designing heuristic reward functions and action selection strategies with harmony functions, they enhanced the effectiveness of AGV motion path exploration [16].

While researchers have achieved some results, there are still the following issues: Firstly, traditional graph-search-based path planning methods, when conducting a complete exploration of the environment, suffer from increased complexity and high computational costs due to the need to traverse all nodes in a directed graph. Moreover, most traditional graph-search-based path planning methods are mainly used for global searches in static environments and cannot meet the needs of dynamic planning scenarios in rail transport environments. Additionally, some traditional path planning methods assume a known environment and do not consider real-world factors when defining multi-robot path planning problems, whereas AGVs in real applications gradually become aware of dynamic environmental information. Secondly, although multi-sensor fusion technology has been used in fields such as vision navigation [17], multi-robot dynamic planning [18], and autonomous driving [10], [19], each multi-sensor fusion technology is based on specific use scenarios (such as obstacle avoidance technology designed for the living scenarios of visually impaired people) and cannot be applied to rail transport environments. Furthermore, different mobile robots are equipped with different types of sensors, and there are significant differences in the processing algorithms for different types of sensors. This article uses AGVs equipped with LiDAR and camera sensors for rail transport environments. In this study, LiDAR (Light Detection and Ranging) sensors are utilized, which emit laser pulses and record the return time of these pulses, enabling precise calculation of the distance distribution in the surrounding environment. LiDAR [20] is not affected by external influences and can better adapt to complex dynamic environments with interference. Using vision sensors [21] can richly perceive environmental information, detect obstacles, and achieve obstacle avoidance. Lastly, most existing multi-agent planning algorithms represent agents with rectangular boxes or other simple shapes. However, this article, based on the rail transport environment, needs to consider the practical application of algorithms. The rail transport material transportation environment is complex, requiring not only consideration of static obstacles but also dynamic obstacles (other AGVs besides oneself). In addition, existing DRL algorithms for path planning are only tailored for specific scenarios, showing inapplicability in the field of rail transportation.

In recent years, path planning has become a critical aspect of research in fields such as mobile robotics and autonomous guided vehicles (AGVs). Currently, traditional approaches, including graph-based methods, predefined-path strategies, and heuristic-based search algorithms, have been widely applied. However, these conventional methods exhibit considerable limitations when dealing with complex and dynamic environments.

Firstly, graph-based path planning methods, such as the A* algorithm [22], [23] and Dijkstra's algorithm [24], typically show limited performance in complex and dynamically changing environments. Although these algorithms are effective in deterministic scenarios, their performance significantly deteriorates when confronting dynamic obstacles. Furthermore, these methods often lack adaptability for real-time navigation tasks, as they must recalculate paths frequently when environmental conditions change [25].

Secondly, fixed-path solutions for AGVs, including cable-guided or magnetic-strip-guided methods [26], provide stable and predictable trajectories, yet suffer from notable drawbacks. Specifically, such approaches necessitate high maintenance costs due to the fixed infrastructure and limited flexibility [27]. Additionally, when encountering unplanned obstacles or congestion scenarios, these methods are unable to adjust paths adaptively, leading to potential efficiency losses.

Moreover, traditional dynamic replanning algorithms, such as D* and its variants (e.g., D* Lite), while capable of local path adjustments in partially unknown environments, still encounter efficiency limitations when dealing with complex scenarios [28], [29]. These methods typically require extensive computational resources to update paths, thereby limiting their real-time application in highly dynamic and complex environments [30], [31].

Given these limitations, researchers have proposed various enhanced methodologies in recent years, including sampling-based Rapidly-exploring Random Trees (RRT) and deep reinforcement learning (DRL) methods, aimed at significantly improving path-planning performance and adaptability in more complex and dynamic operational environments [32].

Addressing the aforementioned issues, this paper focuses on rail transport material handling and primarily investigates multi-AGV path planning methods, which are divided into three main parts as shown in Figure 1. The first part of the research addresses the high time cost issues caused by collisions with obstacles during AGV cargo transportation in complex dynamic rail transport environments. It establishes multi-AGV path planning, combining the characteristics of the rail transport material transportation environment. Using the A* path algorithm as a baseline and integrating traditional path planning methods with deep learning approaches, it autonomously learns heuristic functions to minimize search costs in complex environments. The method is trained by minimizing the squared error between the predicted values at each waypoint and the target cost values, ultimately fitting

each AGV with an optimal path of minimum cumulative cost for global AGV path planning. Traditional AGVs that move along predetermined paths or carry out transportation tasks guided by wires require high maintenance costs for the lines, and AGV queues can cause time costs due to congestion. To address these issues, the second part of the research utilizes the binocular cameras and LiDAR sensors carried by new AGV carts. By capturing the environmental features around the AGVs with sensors and using LSTM networks to extract LiDAR point cloud data and time-based AGV state information for obstacle perception, a twin deep convolutional neural network is employed. Through shared convolutional layers, AGV environmental information perception modules and scene classification modules extract AGV camera information, obtaining classifications of different types of obstacles and targets. The LiDAR information and camera features are integrated with the cart's own information to obtain a comprehensive environmental perception model. In the rail transport material handling environment, multiple AGVs work in parallel, carrying goods and serving as dynamic obstacles to each other, all within a large-scale AGV operating environment. How to achieve collision-free transportation to the destination is very important. For the avoidance of dynamic obstacles (other AGVs, etc.) in a multi-AGV operating environment, the third part of the research proposes a study on multi-AGV dynamic path planning under global guidance based on deep reinforcement learning. Combining the deep reinforcement learning model with global guidance, a differentiated reward and punishment method for multi-AGV dynamic planning is trained: taking the multi-AGV paths output by the deep convolutional model in the first part as global guidance, real-time AGV feedback on motion and environmental information is obtained through the environmental perception model in the second part. The relative positions of waypoints are used to determine short-term goals, and decisions are made using the deep reinforcement learning model, with corresponding rewards and punishments for AGV movements, and each decision-making state is stored in an experience replay pool as a training sample for future use. Different rewards and punishments are given to AGVs depending on the outcome of each navigation attempt. Utilizing the initial path obtained from traditional algorithms as a baseline to provide global guidance, the convergence speed of deep reinforcement learning is accelerated. Through the designed reward function, unmanned vehicles can roughly follow the initial path to quickly reach the target during navigation, with a smoother trajectory and higher navigation efficiency.

II. MATERIALS AND METHODS

A. RESEARCH ON HEURISTIC GLOBAL PATH PLANNING BASED ON WAYPOINT FITTING

There are two types of AGV global path planning from the perspective of application scenarios. One is offline planning, which is aimed at environments that only have static obsta-

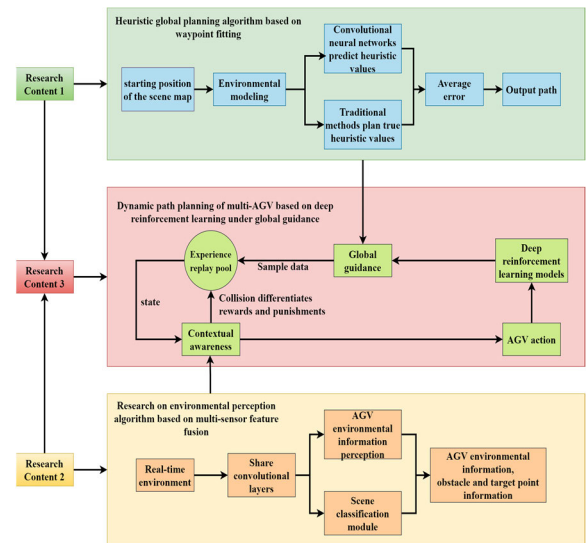


FIGURE 1. Technical roadmap.

cles. This method requires that the AGV knows the entire working environment, meaning the location of each static obstacle is known; the other is online global planning, which is aimed at dynamic working scenarios with partially known environments and dynamic obstacles. The issue with offline planning is that when an AGV encounters an obstacle, it needs to replan, leading to low time efficiency [33]. Both offline and online global planning assume that the environment is largely known. In the context of rail transport logistics, AGVs also face a partially known environment (such as known rail lines). Therefore, this paper takes the traditional global planning algorithm A^* [34] as the baseline, assuming the information of static obstacles in the known environment, and combines deep learning methods to automatically learn heuristic functions to minimize search costs in complex environments. It trains by minimizing the squared error between the predicted values and the cost values of key waypoints (navigational points) on the A^* planned path, fitting an optimal path with the minimum cumulative cost for each AGV, resulting in a multi-AGV global planning model to solve the replanning issue, while also using the model's output of initial paths for multiple AGVs as global guidance in the third part of the research training.

1) MODELING OF AGV WORKING ENVIRONMENT

In the rail transport logistics scenario, Automated Guided Vehicles (AGVs) stand by next to the tracks when trains enter the station [35]. After the train stops, the AGVs start to transport goods from the track to the designated destination. Considering that each station has several parallel tracks, this study adopts a grid method to simulate the rail transport environment. As shown in Figure 2, we designed a 10×12 grid map as a demonstration. In this map, continuous black blocks represent parallel tracks, meaning these areas are impassable due to obstacles. Blue blocks represent AGVs, while the

remaining white space indicates areas where AGVs can move freely.

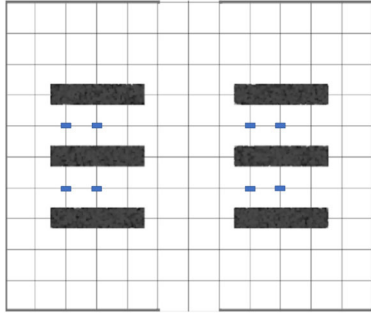


FIGURE 2. Grid diagram of the working environment of AGVs.

2) WAYPOINT FITTING

Convolutional Neural Networks (CNN) possess a remarkable capability to capture both the global structure and detail features of an environment in heuristic learning, generating outputs with spatial structure. This trait aids in the smooth transition of heuristic values between adjacent configurations. Based on this, we apply convolutional neural networks to the heuristic function of the A* path planning algorithm, aiming to enhance the efficiency of path search for multi-AGV systems. As illustrated in Figure 3, this paper addresses the high time cost issue arising from collisions with obstacles requiring replanning during the cargo transportation process by AGVs in complex dynamic rail transport environments. It establishes a multi-AGV path planning task definition, taking into account the characteristics of the rail transport material handling environment. The grid information of the AGV environment and its starting position coordinates are input. Using the A* path algorithm as a baseline, by combining traditional path planning methods with deep learning approaches, it leverages automatically learned heuristic functions to minimize search costs in complex environments. It trains by minimizing the squared error between the predicted values of each waypoint on the path and the target cost values, eventually fitting an optimal path for each AGV with the minimum cumulative cost, achieving global path planning for AGVs. In this process, key waypoints on the path are defined as effective nodes, as shown in Figure 4. These nodes connect the start and end points, forming a set of nodes with the lowest cost, serving as the baseline heuristic true values during the training process.

Define the search graph $G(V, E)$. In this graph, the function $Scc(v)$ is used to expand the search edges, thereby accessing the successor edges and sub-vertices of vertex v . Given that obstacles render some edges impassable, we validate the viability of each potential waypoint through the function $Valid(e, v, \phi)$; this function returns true only if edge e is not occupied by obstacles. Based on the environment ϕ , each candidate waypoint is evaluated using the search score function $Score(v, \phi)$, and then they are added to the priority

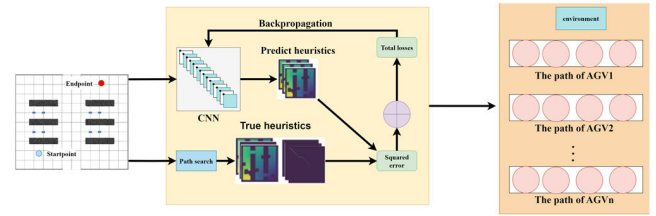


FIGURE 3. Heuristic global programming algorithm based on waypoint fitting.

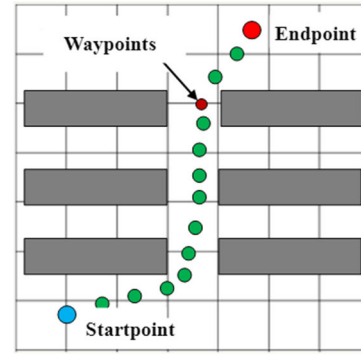


FIGURE 4. Waypoints in the global path.

queue O in order based on their scores. In each iteration, the waypoint with the highest score from queue O is selected, the search scores of its sub-vertices are calculated, and they are added to the queue. This process continues until the target vertex v_g is reached or the queue O is empty. The path cost $g(v, \phi)$ is calculated by accumulating the shortest path edge costs $Cost(e, v', \phi)$ encountered during the search process. As defined by (1), the search heuristic function $h(v, \phi)$ assigns an integrated score to each waypoint to balance efficiency and accuracy. This study, targeting environments containing obstacles, extracts feature maps and applies a Fully Convolutional Neural Network (FCNN) [36] to predict the heuristic value of each waypoint. Training is conducted by minimizing the squared error between the predicted value of each waypoint and its actual target cost value. The cost of a waypoint is defined as the cumulative cost from that point to the target point along the shortest path. Training is completed by minimizing the objective function defined in (2), with the goal of accurately predicting waypoint costs to optimize path planning. In the model, $\hat{H}$ represents the target cost value, and M acts as a mask function, assigning a value of 1 to each vertex on the path, to exclude invalid vertices, such as those occupied by obstacles or completely surrounded areas, when generating target values. To address the lack of supervisory signals for unvisited pixels, this study adopts a temporal difference method to compensate for the insufficient supervisory signals, as specifically defined in (3). Predicted values $h(v, \phi)$ are iteratively updated through convolution with fixed kernels and biases, minimizing across the successor vertex axis, with the loss function defined in (4). Here, the value of $M_{TD}(v)$ is 1 when $v \in \bar{M} \cap V_{valid}$, otherwise, it is 0, and λ represents

the weight coefficient for the temporal difference loss. This method refines the target cost estimation through multiple iterative updates. Training aims to plan the optimal global path for Automated Guided Vehicles (AGVs) by minimizing the squared error between the predicted values and the true target costs.

$$\text{Score}(v, \phi) = g(v, \phi) + h(v, \phi) \quad (1)$$

$$L(\phi, \hat{H}, M) = \sum_{v_i \in V} \left((h(v, \phi) - \hat{H}(v))^2 M(v) \right) \quad (2)$$

$$\hat{h}(v, \phi) \leftarrow \min_{(e, v') \in \text{Sec}(v)} \text{Cost}(e, v', \phi) + \hat{h}(v', \phi) \quad (3)$$

$$l(\phi, \hat{H}, M) = \sum_{v \in E} \left\{ \left((h(v, \phi) - \hat{H}(v))^2 M(v) + \lambda (h(v, \phi) - \hat{h}(v))^2 \right) M_{TD}(v) \right\} \quad (4)$$

3) CONFLICT RESOLUTION

In a multi-agent environment composed of N Automated Guided Vehicles (AGVs), the starting position v_s and the target position v_g of each AGV are connected through a series of waypoints, defining the waypoint path set as $p = \{r^k\}_0^{N_p}$. Here, $r^k = \{x^k, y^k, \theta^k\}$ represents the k th waypoint on the path, where N_p is the total number of waypoints, and (x^k, y^k) and θ^k respectively represent the two-dimensional position and direction of the AGV.

In the process of planning paths for each AGV individually, considering that all AGVs share the same graph $G(V, E)$, this study identifies and addresses two potential types of conflicts: vertex conflicts and edge conflicts. A vertex conflict occurs when two AGVs, Agent_i and Agent_j , occupy adjacent vertices at any time step t , denoted as $\langle \text{Agent}_i, \text{Agent}_j, t \rangle$. Similarly, an edge conflict occurs between two consecutive time steps t_1 and t_2 if two AGVs come too close while traversing an edge, denoted as $\langle \text{Agent}_i, \text{Agent}_j, t_1, t_2 \rangle$. Here, “too close” is defined by a collision model with a radius R centered on the AGV’s central coordinates. The set of vertex conflicts at time step t is described by (5). The set of edge conflicts at time steps (t_1, t_2) is described as (6).

$$\text{VCon}(t) = \{(i, j) | R(\text{pos}(r_i^t)) \cap R(\text{pos}(r_j^t)) \neq \emptyset\} \quad (5)$$

$$\text{ECon}(t_1, t_2) = \{(i, j) | R(\text{pos}(r_i(t_1))) \cap R(\text{pos}(r_j(t_2))) \neq \emptyset, \forall t \in (t_1 \Delta T, t_2 \Delta T)\} \quad (6)$$

In a multi-AGV system, the function $\text{pos}(r_i(t))$ defines the two-dimensional coordinates of the waypoint r_i for agent Agent_i at time t , where R represents the specific position (x, y) of the AGV in the environment and the subset of the environment it occupies. By checking the global path solutions in time order, we determine whether conflicts exist. Assuming a vertex conflict $\langle \text{Agent}_i, \text{Agent}_j, t \rangle$ is detected, a method of constructing a binary constraint tree is adopted to generate two child nodes aimed at resolving the conflict. Specifically, in the first child node, a constraint is applied to Agent_i to prevent it from being at position r_i^t at time

t ; for the second child node, a corresponding constraint is applied to Agent_j to prevent it from being at position r_j^t at time t . A similar strategy is adopted to handle edge conflicts $\langle \text{Agent}_i, \text{Agent}_j, t_1, t_2 \rangle$, and the search continues until a conflict-free solution is obtained.

4) TRAJECTORY OPTIMIZATION

In this section, we optimized the obtained discrete paths, converting them into smooth trajectories. Specifically, a time stamp $t^l = l \Delta t$ is assigned to each waypoint in the global path $p = \{r^t\}_0^H$ for each Automated Guided Vehicle (AGV). Thus, the global path is transformed into a reference trajectory $v^r = \{r^t, t^l\}_0^H$, where each segment on the original global path is evenly divided into h parts. At the same time, the kinematic model for each AGV is defined as (7).

$$\dot{Z} = F(z, u), u \in U \quad (7)$$

In this framework, the state z of an AGV is defined as its two-dimensional position and orientation $[x, y, \theta]^T$, $u = [v, w]^T$. Here, v and w respectively represent the AGV’s linear and angular velocities. The optimal smooth trajectory is defined as the set $v = \{z^t, t^l\}_0^H$, and the corresponding nonlinear optimization problem can be formalized as (8).

$$\text{minimize} \sum_{i=1}^N \left(\sum_{j=1}^{H-1} \Delta u_i^T P \Delta u_i + \sum_{j=1}^H \hat{z}_i^T Q \hat{z}_i \right) \quad (8)$$

where, $\Delta u_i^j = u_i^j - u_i^{j-1}$ and $\hat{z}_i^j = z_i^j - r_i^j$. Here, $P \in \mathbb{R}^{2 \times 2}$ and $Q \in \mathbb{R}^{3 \times 3}$ are two positive definite weight matrices, respectively. Formula (8) must satisfy conditions such as (9)-(13).

$$z_i^0 = v_{s_i}, z_i^H = v_{g_i} \quad \forall i \quad (9)$$

$$z_i^{j+1} = F(z_i^j, u_i^j), \quad \forall i, j \quad (10)$$

$$\text{pos}(z_i^j) \in S_i^j, \quad \forall i, j \quad (11)$$

$$u_i^j \in U_i, \quad \forall i, j \quad (12)$$

$$\| \text{pos}(z_{i_1}^j) - \text{pos}(z_{i_2}^j) \| \geq R_{i_1} + R_{i_2}, \quad \forall i_1, i_2, j, i_2 \neq i_1 \quad (13)$$

In this framework, S_i^j represents the safety space considered when extending the global path. According to Formula (8), the objective function contains two main parts. The first part promotes the smoothness of the trajectory by imposing a penalty on the differences between consecutive control inputs. The second part considers that the reference global trajectory v^r of each Automated Guided Vehicle (AGV) already represents a feasible solution. To maintain the feasibility of the optimization problem, a penalty is imposed on the deviation between the optimal trajectory and the reference global trajectory.

At each time step t , the state transition is given by $z_i^{j+1} = F(z_i^j, u_i^j)$, where the nonlinear model F is its discrete-time version. The position of each waypoint on the trajectory is constrained by the safety space which is settled by constraint

condition $\text{pos}(z_i^j) \in S_i^j$. In $u_i^j \in U_i$ the control input for each AGV, is limited by the range of its physical capabilities. Additionally, the requirement to maintain a safe distance between each pair of AGVs is represented by $\| \text{pos}(z_{i_1}^j) - \text{pos}(z_{i_2}^j) \| \geq R_{i_1} + R_{i_2}$.

Trajectory optimization primarily focuses on the constraints between AGVs (constraint 8f). Since the optimal trajectory approaches the reference trajectory according to the cost function (8a), the constraints between AGVs can be analyzed based on the reference global trajectory $v_1^r, \dots, v_N^r$. At each time step t , a search for constraint satisfaction is conducted by analyzing the combination of three AGVs $(i_1, i_2, i_3)^t$ with the aim of satisfying (14).

$$\| \text{pos}(r_a^t) - \text{pos}(r_b^t) \| \leq D_{th}, \forall a, b \in \{i_1, i_2, i_3\}, b \neq a \quad (14)$$

In this section, we introduce a threshold $D_{th} = \sqrt{2}D$ to judge the proximity of three AGVs to each other within a specific time step, where each triplet represents the closeness situation of three AGVs at that time step. In the discrete path planning stage, we can only confirm that four AGVs simultaneously satisfy the condition of (14) when the four AGVs are located at the four vertices of the same grid cell.

When AGVs in a triplet are assigned different priorities, the AGV with higher priority will plan its trajectory first. Subsequently, these trajectories will act as hard constraints for the optimization problem of AGVs with lower priority. In this case, the original problem may no longer have a feasible solution. Therefore, it is suggested to consider the AGVs within a triplet as a whole and assign them the same priority. A list L is defined to represent all triplets that satisfy the condition of (14).

If the number of AGVs in the workspace is large, L will contain a large number of elements, making the same AGV appear in multiple triplets. In such cases, it is not straightforward to group AGVs. Each triplet represents a scenario where three AGVs are close to each other, and if the same triplet appears multiple times in L , it indicates that the trajectories of these three AGVs are highly coupled. Based on this, we propose an algorithm to assign priorities based on the frequency of triplets' occurrences.

Initially, the element with the highest frequency of appearance in L is designated as the first priority group. Since the AGVs in this new group may appear in other triplets in L , all these AGVs will be removed from L , generating a new list L' . Consequently, the elements in L' may contain three, two, or just one AGV. Next, the element with the highest frequency of appearance in the new list L' is chosen as the second priority group, and this process is repeated until the list is empty. If an AGV does not have significant coupling with other AGVs, it will not appear in the original list L , and such AGV is considered as a separate AGV group, assigned the lowest priority.

Subsequently, following the order of priority from high to low, the trajectory optimization problem for each AGV group is formulated and solved. Figures 5-a to 5-d illustrate

examples of the overall priority-based trajectory optimization process. In each iteration, the current AGV group needs to avoid all AGVs with higher priority. The trajectories of the current AGV group and the higher priority AGVs are represented by colored lines and black lines, respectively. For instance, first, 8 AGVs are grouped into four priorities, namely $([(2, 4, 5), (7, 8), (3, 6), (1)])$. In each iteration of priority trajectory optimization, in addition to considering the constraints within the current group of AGVs, the constraints between the current AGV group and the higher priority AGV groups are also taken into account to ensure that no collisions occur.

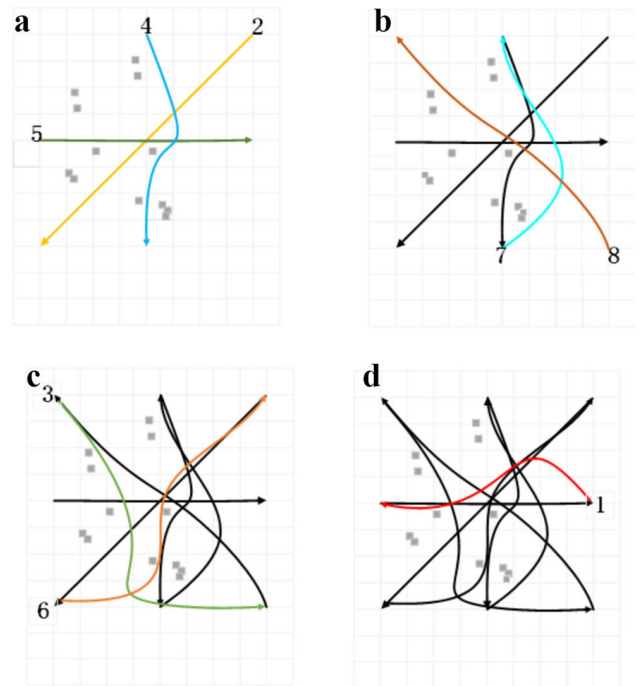


FIGURE 5. Priority track optimization example.

5) EXPERIMENTAL RESULTS AND ANALYSIS

a: MULTI-AGV GLOBAL PATH PLANNING RESULTS

In this study, we simulated a material transportation system within a subway traffic environment, providing a detailed description of the subway track layout and its surrounding environment. As illustrated in Figure 6, the subway tracks are represented by regularly arranged continuous gray dots, while static obstacles are identified by scattered gray dots around them. Based on this, we placed several Automated Guided Vehicles (AGVs), with their starting positions distributed around the tracks and in relatively distant areas. These AGVs are distinguished by rectangles of different colors, and their carrying trajectories are represented by long lines of different colors. Through this layout, we demonstrate how each AGV successfully navigates around obstacles to reach its randomly assigned target location, with the AGV's final docking point marking its target position.

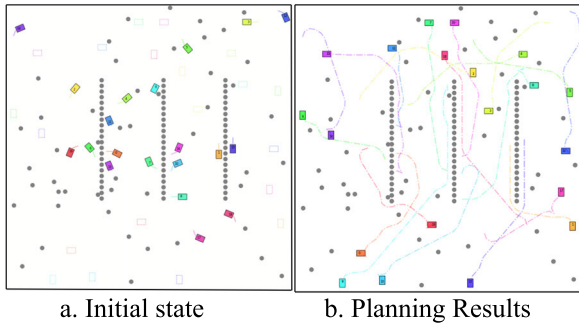


FIGURE 6. Global path planning trajectory diagram for multiple AGVs.

b: CONFLICT RESOLUTION

Further, Figure 7 reveals the dynamic process of conflict resolution in a multi-AGV system. By marking the AGVs around the imminent conflict point with red and black circles respectively, we observed that at $t = 3$ seconds, two AGVs approached the conflict point. As time progressed, the purple AGV detected a potential conflict at $t = 4$ seconds and chose to temporarily wait, allowing the orange AGV to pass first at $t = 5$ seconds. Once $t = 6$ seconds passed and the orange AGV safely went through, the purple AGV continued its path, successfully avoiding the conflict. Another scenario occurred at $t = 8$ seconds, where the orange AGV and the pink AGV, marked with a blue circle, were about to conflict. The pink AGV avoided the conflict with the orange AGV by executing a quick retreat at $t = 10$ seconds.

c: TRAJECTORY OPTIMIZATION RESULTS

Moreover, Figure 8-a showcases the results of trajectory optimization based on modeling the AGV working environment. By smoothing the trajectories of discrete paths, we positioned some AGVs around the tracks as starting points, while others are placed outside the tracks, acting as temporary standby AGVs. Ultimately, Figure 8-b presents a three-dimensional visualization of the optimized trajectories, intuitively demonstrating how AGVs efficiently navigate around obstacles in a complex environment to reach their predetermined targets.

B. RESEARCH ON ENVIRONMENTAL PERCEPTION ALGORITHM BASED ON MULTI-SENSOR FEATURE FUSION

In the orbital transportation environment, Automated Guided Vehicles (AGVs) are responsible for tasks such as cargo handling. Given the limited dwell time of trains at each station, AGVs need to complete tasks quickly and be capable of real-time obstacle avoidance. To enhance transportation efficiency, AGVs must possess autonomous path planning capabilities. This study utilizes the built-in sensor equipment of AGVs to perceive the surrounding environment and their own state, assisting AGVs in making intelligent decisions to cope with the complexity and variability of the environment. This enables AGVs to plan an optimal path from the starting point to the destination in a complex environment filled with dynamic obstacles.

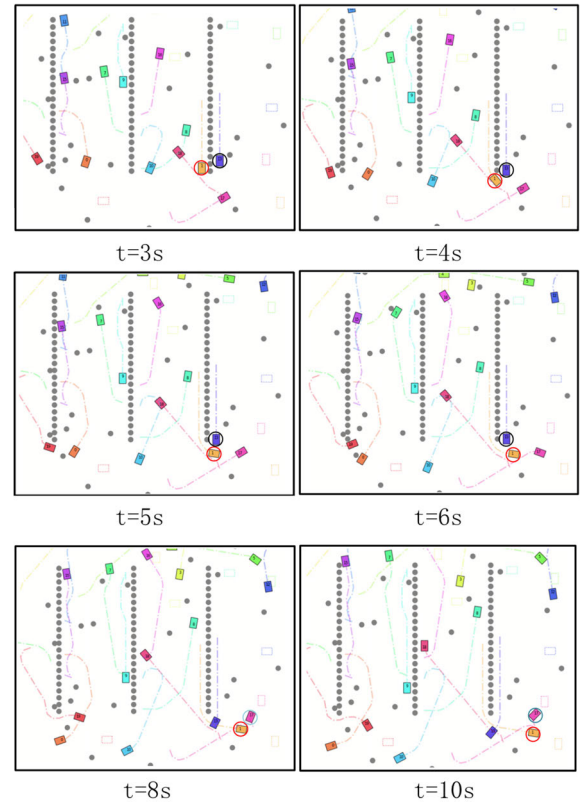


FIGURE 7. Multi-AGV path conflict resolution diagram.

1) VISUAL FEATURE PROCESSING BASED ON DUAL MODULES

In modern automated logistics systems, traditional Automated Guided Vehicles (AGVs) rely on predetermined paths or are guided by ground wires to perform handling tasks [37]. However, this reliance on highly maintained path planning not only increases significant maintenance costs but also leads to queue congestion among AGVs during task execution, resulting in reduced efficiency. To address these issues, this study developed a comprehensive environmental perception model that integrates point cloud data generated by LiDAR, the status information of AGVs, and visual images captured by front-mounted cameras. The model uses deep reinforcement learning techniques to provide extensive perception capabilities for the path planning of multiple AGVs.

In this study, the AGV is equipped with two primary sensors: a LiDAR (Light Detection and Ranging) sensor and a front-mounted camera. The LiDAR sensor selected is the fictional model LaserSense-16, which provides a full 360° field of view with a detection range of 0.5 to 50 meters, a horizontal angular resolution of 0.2° , a vertical angular resolution of 2° , and a data refresh rate of 10 Hz. This sensor accurately captures the distance information of surrounding obstacles, effectively supporting the construction of real-time environmental models. The front-mounted camera, a fictional model VisionPro-HD, features a resolution of 1920×1080 pixels

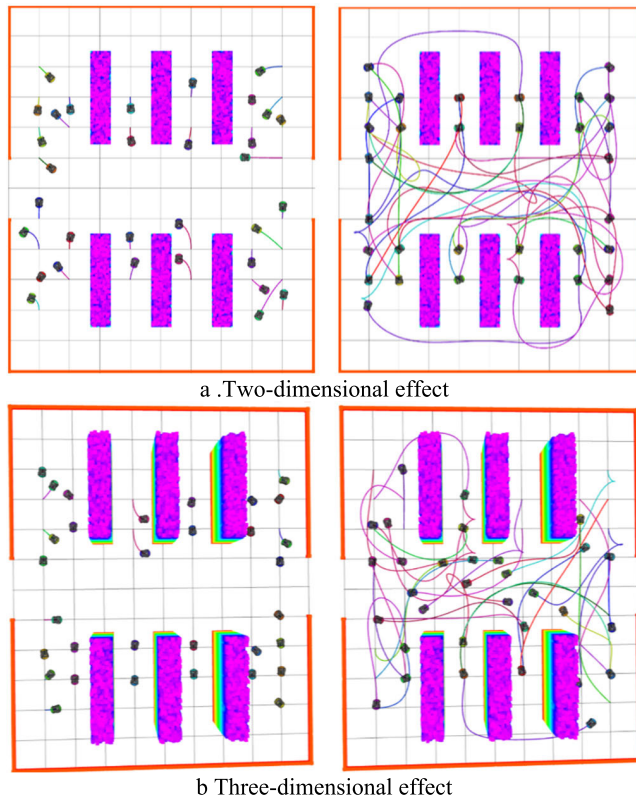


FIGURE 8. Optimization effect of multi-AGV global path planning.

and a frame rate of 30 FPS, enabling the capture of rich visual data primarily used for obstacle recognition and target detection tasks. In subsequent research, multi-sensor data fusion techniques will be employed to integrate LiDAR and camera data, further improving the AGV's environmental perception accuracy and path-planning performance.

Figure 9 presents the proposed multi-sensor data fusion-based environmental perception framework of this study, aimed at enhancing the accuracy and response speed of AGVs' perception of the surrounding environment. As shown in Figure 9-a, the simulated AGV equipped with cameras for real-time obstacle avoidance is demonstrated. For this purpose, it is necessary to train the AGV so that it can accurately perform scene classification based on the visual information captured by the camera, identifying other AGVs, static obstacles, and target points. The goal is to ensure that the AGV can reach the specified destination smoothly without collisions. The image resolution captured by the AGV's front-mounted camera is $768 \times 1024 \times 3$. To improve the efficiency of path planning in obstacle avoidance training while retaining image content, the original images are processed into grayscale images of $80 \times 80 \times 3$ before entering the shared convolutional layers.

We designed and implemented a multi-sensor data processing framework that adopts a dual-branch structure, sharing input data and several convolutional layers. As shown in Figure 10, we utilized a Siamese Deep Convolutional Neu-

ral Network (DCNN) architecture, which processes inputs from both the AGV environmental information perception module and the scene classification module simultaneously through shared convolutional layers, further extracting image information captured by the AGV camera. This approach enables us to distinguish between different types of obstacles and targets, and integrates LiDAR and camera features with the AGV's own information to form a comprehensive environmental perception model. As shown in Figure 10-a, the shared convolutional layers, consisting of multiple convolutional and max pooling layers, aim to extract key features from the input images, serving as input for the AGV's environmental perception and scene classification modules. The purpose of this design is to enhance the model's processing efficiency and achieve efficient memory usage and reduced inference time, without compromising the prediction accuracy of the two branches. By integrating the AGV's environmental information and scene semantic segmentation, the framework achieves precise localization of the AGV and effective classification of the visual information obtained through the front-mounted camera. The architecture of the AGV environmental information perception module, as shown in Figure 10-b, includes three convolutional layers, two max pooling layers, two global average pooling layers, and one fully connected layer. This module aims to precisely locate the AGV within the captured scene, especially determining the AGV's orientation angle and position relative to the current waypoint, as illustrated in Figure 10-d. Through the environmental information perception module, the AGV's current self-information (dt, ct) can be obtained, where dt represents the distance between the AGV and the nearest waypoint on the global path, and ct indicates the orientation angle of the AGV relative to that waypoint. As demonstrated in Figure 10-c, Atrous Spatial Pyramid Pooling (ASPP) technology is employed. This technique captures object and image context information at multiple scales using dilated convolutional layers with different sampling rates. This method allows the model to integrate multi-scale features into its effective receptive field, thereby reconstructing dense features from the sparse outputs generated by the network. As shown in Figure 10-e, this module aims to achieve pixel-wise segmentation of the incoming scene, subsequently classifying the scene content by labels: dynamic carts, static obstacles, and target points. Specifically, dynamic obstacles are marked in red, static obstacles in blue, and target points in green for visual distinction.

2) LIDAR INFORMATION PROCESSING

In this study, the simulated Automated Guided Vehicle (AGV) is equipped with LiDAR and generates point cloud data. Due to the long sequence characteristic of point cloud data over time steps, Long Short-Term Memory (LSTM) networks are used for processing, thus extracting distance and angle information about target points and obstacles, as shown in Figure 10-b. The LSTM network starts from an initial empty state, receiving the observation state S_2^I at the first time

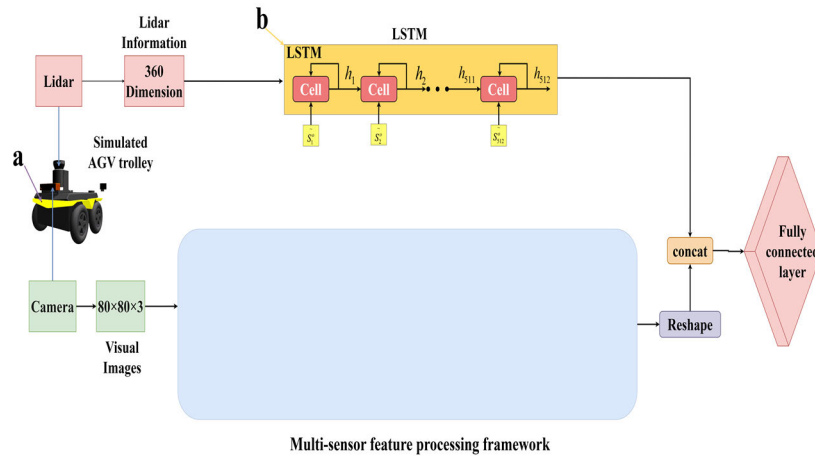


FIGURE 9. An environment-aware framework based on multi-sensor feature fusion.

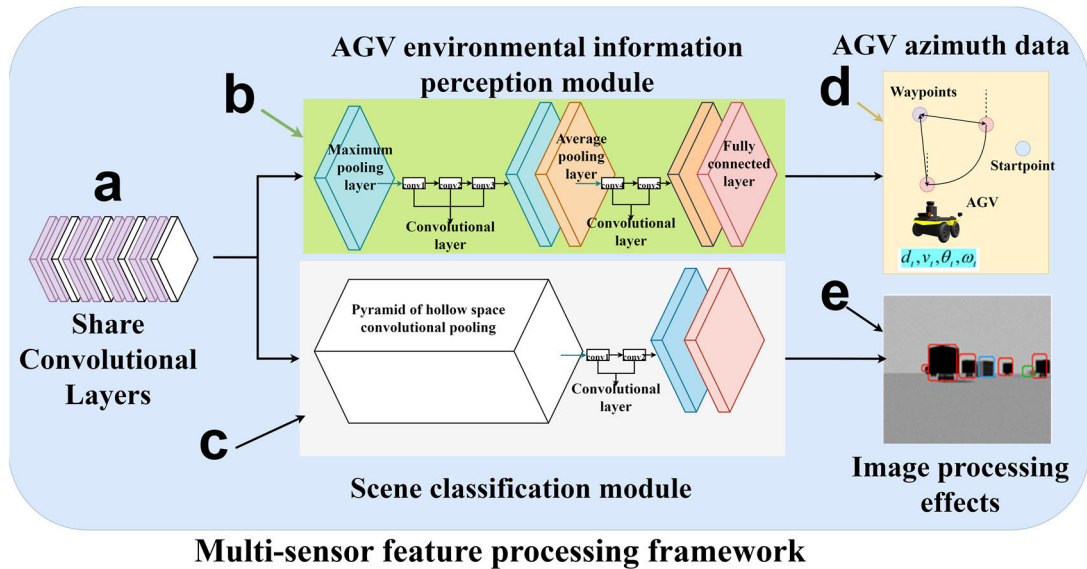


FIGURE 10. Multi-sensor feature processing framework.

step to generate the hidden state vector h_2 . Thereafter, the network generates a new hidden state h_n based on h_1 and S_2^l , repeating this process until the AGV reaches its destination. In this study, the LSTM network's cell units are set to 512 to handle the 360-degree point cloud information output by the LiDAR, with an update frequency set at 50Hz and a detection range of $[-90^\circ, 90^\circ]$. Considering that the AGV's own information (such as linear velocity v_t and angular velocity ω_t) is also based on time steps, this study also employs LSTM networks to process both the LiDAR point cloud information and the AGV's linear and angular velocities v_t and ω_t , to achieve more accurate environmental perception and motion control. As the AGV moves from its starting position to its destination, the environmental perception module can obtain and process the state information of the surrounding environment in real-time, providing a real-time perception field of view for the AGV. Figure 11 shows an example of the real-time perception

field of view captured by the AGV through the environmental perception module, demonstrating how the AGV can effectively identify and respond to environmental changes along its path.

C. RESEARCH ON DYNAMIC PATH PLANNING OF MULTI-AGV BASED ON DEEP REINFORCEMENT LEARNING UNDER GLOBAL GUIDANCE

Traditional multi-agent planning algorithms typically represent agents with simple shapes, such as rectangular boxes, and most of these algorithms are only applicable to static and fully known planning scenarios. However, this study focuses on the operational environment of AGVs (Automated Guided Vehicles) in the rail transit domain, where not only static obstacles need to be considered but also dynamic obstacles, such as other AGVs. Moreover, existing AGV path planning algorithms in domains like warehouse logistics are not

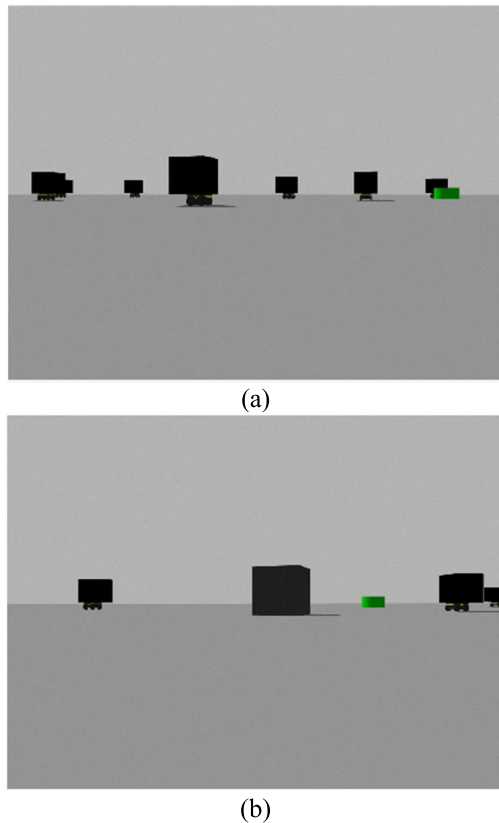


FIGURE 11. Real-time capture of environment awareness.

suitable for the rail transit environment, demonstrating their limitations.

This study proposes an innovative AGV path planning method for rail transit applications. By designing novel reward functions and incorporating deep reinforcement learning, we allow AGVs to learn through interaction with the environment without relying on predefined training sets. Although this process may face challenges such as delays and noise, it still offers significant advantages over traditional multi-agent planning algorithms. This paper particularly emphasizes the integration of global guidance and multi-sensor feature fusion techniques within the deep reinforcement learning framework, to accelerate the model's convergence rate and effectively achieve dynamic obstacle avoidance among multiple AGVs, optimizing path planning.

For static obstacle detection, this study adopts a one-time global planning strategy, ensuring that AGVs can reach their target locations via an obstacle-free path. In contrast, when facing dynamic obstacles such as other AGVs, this method predicts their motion trajectories to assess collision risks. While the artificial potential field method [38], the extended Rapidly-exploring Random Trees (RRT*) [39], and the dynamic D* algorithm [34] are typical methods for solving planning problems in dynamic environments, each has its limitations. For example, the artificial potential field method is prone to getting stuck in local minima, and the D* algorithm is inefficient when dealing with changes in nodes far from the AGV.

1) MULTI-AGV DYNAMIC PATH PLANNING AND DESIGN

In the rail transit material handling environment, multiple AGVs work in parallel, transporting goods and serving as dynamic obstacles to each other. In environments where a large number of AGVs operate simultaneously, it is crucial to achieve collision-free delivery to the target points. Addressing the avoidance of dynamic obstacles (such as other AGVs) in a multi-AGV operational environment, as illustrated in Figure 12, this paper presents a study on multi-AGV dynamic path planning under global guidance based on deep reinforcement learning. Integrating a deep reinforcement learning model with global guidance, a differentiated reward and punishment method for multi-AGV dynamic planning is trained: the multi-AGV paths output by the deep convolutional model from the first part of the study serve as global guidance. Through the environmental perception model detailed in the second part of the study, real-time motion and environmental information fed back by AGVs are obtained to determine the relative positions of waypoints and set short-term goals. Combined with decision-making by the deep reinforcement learning model, corresponding rewards or punishments are applied to AGVs' movements, and the state after each decision is stored in the experience replay pool as a training sample for future use. Different rewards or punishments are given to AGVs based on the outcomes of each navigation attempt. Using initial paths obtained by traditional algorithms as a baseline to comprehensively guide the AGVs' deep reinforcement learning accelerates the convergence speed of the deep learning model. Through a custom-designed reward function, autonomous vehicles can approximately follow the initial paths to quickly reach their targets, achieving relatively smooth trajectories and high navigation efficiency.

In considering the path planning for Automated Guided Vehicles (AGVs), this study treats other AGVs as dynamic obstacles and employs the Deep Q-Network (DQN) [40] as the framework for reinforcement learning (RL). This paper has developed a multi-AGV path planning framework based on deep reinforcement learning, which operates under the guidance of a global planner to optimize AGVs' dynamic obstacle avoidance strategies. As illustrated in Figure 12, the framework perceives environmental states through deep learning (DL) techniques and decides the AGV's action strategy a_t based on this. Whenever an AGV executes an action, it enters a new state s_{t+1} , and receives a reward or punishment according to a specific reward function r_t , thereby generating a new sample (s_t, a_t, r_t, s_{t+1}) . These samples are collected into an experience pool, providing historical data support for subsequent training. Through this method, this study aims to achieve efficient path planning and obstacle avoidance for AGVs in complex dynamic environments, reinforcing their autonomous navigation capabilities. The utilization of the experience pool ensures that the model can learn from past interactions and accelerate the learning process and improve decision quality through iterative improvements.

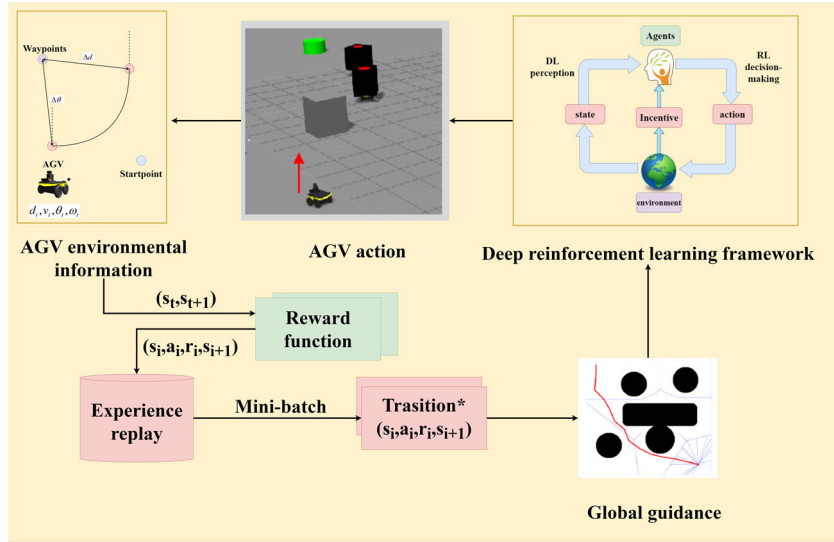


FIGURE 12. A multi-AGV path planning framework based on deep reinforcement learning under global guidance.

2) REWARD FUNCTION DESIGN

Based on the AGV's own environmental information parameters obtained by the AGV environmental information perception module, and after the introduction of global guidance, the target point in the reward function is set as the nearest waypoint. As shown in Figure 13, this study defines the relative distance and heading deviation angle between the Automated Guided Vehicle (AGV) and the target waypoint at continuous time steps t and $t + 1$ as d_{gt} , d_{gt+1} and θ_{gt} , θ_{gt+1} , respectively. Meanwhile, the relative distance and heading deviation angle between the AGV and obstacles at corresponding time steps are defined as d_{ot} , d_{ot+1} and θ_{ot} , θ_{ot+1} .

$$r_t = \begin{cases} K_1 \cdot (d_{gt+1} - d_{gt}) + K_2 \cdot (\theta_{gt+1} - \theta_{gt}) + K_3 \cdot (d_{ot+1} - d_{ot}) + K_4 \cdot (\theta_{ot+1} - \theta_{ot}) \\ K_5 \\ K_6 \end{cases} \quad (15)$$

The design of the reward function in this study is based on the criteria of (15). When an AGV collides with an obstacle, the reward function $r_t = K_5$; when the AGV reaches within 0.5 range of the target point, $r_t = K_6$; under other conditions, the reward function $r_t = K_1 \cdot (d_{gt+1} - d_{gt}) + K_2 \cdot (\theta_{gt+1} - \theta_{gt}) + K_3 \cdot (d_{ot+1} - d_{ot}) + K_4 \cdot (\theta_{ot+1} - \theta_{ot})$, which is determined by the relative distance between the AGV and the target, the change in heading deviation, as well as the distance between the AGV and obstacles, and the change in heading deviation.

The reward function has been carefully designed to guide Automated Guided Vehicles (AGV) to efficiently plan paths and avoid obstacles. The reward factor K includes several parts, as follows: K_1 represents the reward induced by changes in the distance between the AGV and the target. When the distance to the target decreases, K_1 provides a positive reward, motivating the AGV to move closer to the

target. K_2 is based on changes in the heading deviation angle between the AGV and the target. A decrease in the heading deviation angle means that the AGV is moving in the correct direction, and therefore, K_2 rewards this positive change. K_3 and K_4 reflect the rewards associated with changes in the relative distance and course angle to obstacles, respectively. Specifically, K_3 is a negative value, designed to increase the penalty when the AGV is too close to an obstacle, thereby encouraging the AGV to maintain a safe distance. K_4 , on the other hand, rewards the AGV for making course adjustments to avoid obstacles. Significant penalties K_5 and large rewards K_6 are respectively applied when the AGV collides with an obstacle or successfully reaches the target, marking the end of the current Episode and initiating a new learning process. Through this reward mechanism, this study aims to optimize the AGV's path planning strategy, achieving effective obstacle avoidance and target navigation in complex environments.

D. COMPARATIVE ANALYSIS OF PERFORMANCE

To more intuitively and objectively verify the effectiveness and superiority of the proposed method in dynamic path planning and obstacle avoidance for multiple AGVs, five standard benchmark test environments with dimensions of 30 m $\times$ 30 m were established, as shown in Figure 14. In each scenario, ten pairs of start and goal points were randomly generated, with the specific start and goal coordinates for the primary AGV listed in Table 1, ensuring the diversity and representativeness of the test conditions.

By comparing the trajectory results generated by the proposed method (Fig 15(a)) and the traditional Artificial Potential Field (APF) method (Fig 15(b)), clear performance differences can be observed. Fig 15(a) shows the trajectory obtained by the proposed method, where the primary AGV's path is highlighted in red. It demonstrates that the proposed

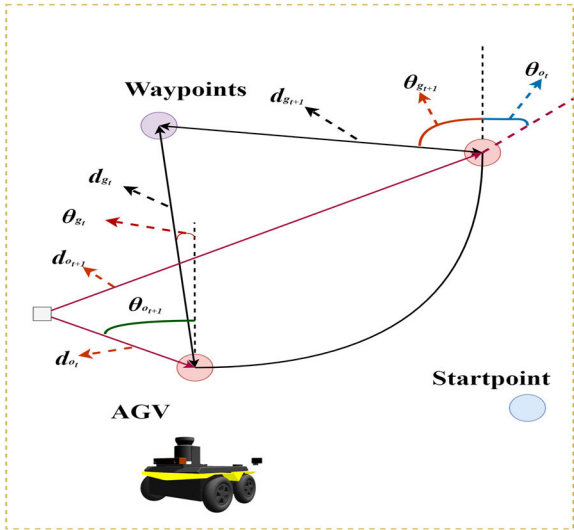


FIGURE 13. AGV environment information.

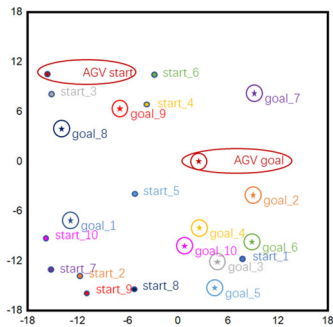


FIGURE 14. Schematic diagram of any starting point and target point.

TABLE 1. Starting point coordinates of the main AGV.

Map	Starting point coordinates (m)	Target point coordinates (m)
1	(12.34, -10.67)	(-0.53, 6.06)
2	(11.16, 14.30)	(-13.68, -3.08)
3	(0.66, 14.81)	(-6.28, -6.33)
4	(-2.50, -12.01)	(3.88, 11.30)
5	(-9.88, -11.57)	(-4.19, 14.33)

algorithm can effectively predict the movement of dynamic obstacles and proactively adjust the path in advance, resulting in shorter and smoother trajectories. In contrast, as illustrated in Fig 15(b), the APF method lacks anticipatory perception of dynamic obstacles, which causes frequent temporary detours and substantial fluctuations in trajectories, significantly increasing the path length and reducing smoothness.

This phenomenon primarily occurs because the APF method relies solely on real-time calculations of attraction and repulsion forces, making it susceptible to falling into local minima and resulting in path instability, particularly in dense dynamic obstacle scenarios.

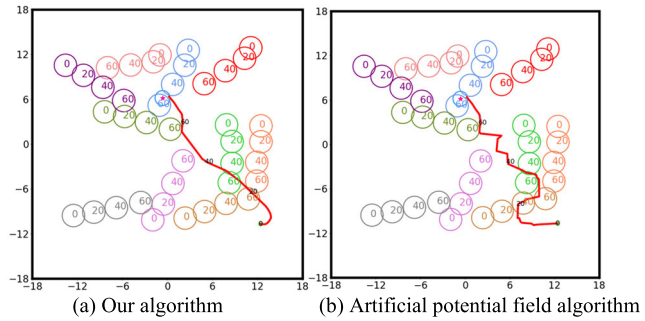


FIGURE 15. Trajectory diagrams of the algorithm and artificial potential field algorithm in this article.

To quantitatively evaluate and compare the path planning performance of the two methods, Fig 16 summarizes the quantitative comparison results in terms of obstacle avoidance capability, computation time, shortest path length, and trajectory smoothness. Fig 16(a) compares obstacle avoidance capability based on the collision count. The results clearly demonstrate that the proposed method consistently yields fewer collisions across all five benchmark maps (Map 1–Map 5), with an average collision reduction of approximately 70% compared to the APF method, highlighting its superior dynamic obstacle avoidance performance and significantly improved system safety and stability. Fig 16(b) illustrates the comparison results for path planning computation time, showing that the proposed method significantly reduces the average computation time by approximately 25%–30% compared to the APF method. These results suggest that the proposed method is more advantageous for real-time path planning applications, rapidly generating feasible and efficient paths in complex dynamic environments. Fig 16(c) presents the shortest path length comparison, indicating that the proposed method consistently generates shorter paths across all scenarios, reducing the average path length by approximately 10%–15%. These results further confirm the outstanding performance of the proposed algorithm in path optimization by effectively avoiding local oscillations commonly seen in APF, thus enhancing direct navigation efficiency and reducing energy consumption for AGVs. Fig 16(d) shows the comparison results for trajectory smoothness, evaluated by the Root Mean Square (RMS) curvature. It is evident that the trajectories generated by the proposed method exhibit significantly lower curvature, with an average reduction of approximately 20%–25%, effectively reducing oscillations and abrupt turns in the paths. Smoother trajectories reduce vehicle vibrations and improve the stability and control accuracy of the AGVs.

In summary, comprehensive analysis from Fig 14 to Fig 16 clearly indicates that the proposed method demonstrates

substantial overall performance advantages over the traditional Artificial Potential Field method. These advantages mainly stem from the combination of deep reinforcement learning techniques and multi-sensor fusion-based environmental perception models proposed in this study, enabling efficient path planning and dynamic obstacle avoidance in complex dynamic environments. Benchmark experiments fully validate that the proposed method significantly enhances key performance indicators such as obstacle avoidance capability, computational efficiency, path length reduction, and trajectory smoothness. Therefore, the path planning algorithm developed in this study holds significant application potential for rail transportation logistics and intelligent AGV navigation systems, particularly in scenarios requiring efficient path planning in complex dynamic environments.

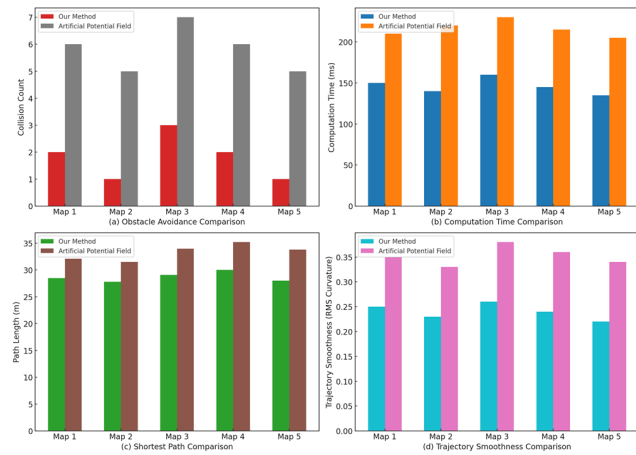


FIGURE 16. Performance comparison in obstacle avoidance, efficiency, path length, and smoothness.

E. EXPERIMENTAL COMPARISON USING THE MAZE BENCHMARK MAP

To further validate the generalization capability and robustness of the proposed dynamic path planning method under standardized testing scenarios, this study introduces the publicly available Maze Benchmark Map for performance comparison experiments. Characterized by its complex structure and dense obstacle distribution, this benchmark is widely employed to evaluate the optimality and convergence capability of path planning algorithms in static obstacle environments, and is considered highly representative.

As illustrated in Figure 17, the experimental environment is constructed using a 50×50 maze grid map, where gray blocks denote static obstacles and the blank areas represent navigable space. The start point (S) and goal point (G) are positioned at opposite corners of the diagonal, forming a typical and challenging global path planning task.

To comparatively evaluate the planning performance of different algorithms in static environments, three representative methods are selected for experimental analysis: (1) the proposed GG-DRL (Global-Guided Deep Reinforcement

Learning) method, which integrates a global guidance mechanism with a multi-sensor feature fusion deep reinforcement learning model; (2) the DDPG [42] (Deep Deterministic Policy Gradient) algorithm, a classical model-free deep reinforcement learning approach designed for continuous action spaces; and (3) the RRT (Rapidly-exploring Random Tree) algorithm, a representative of traditional path planning methods known for its high exploration efficiency in high-dimensional spaces.

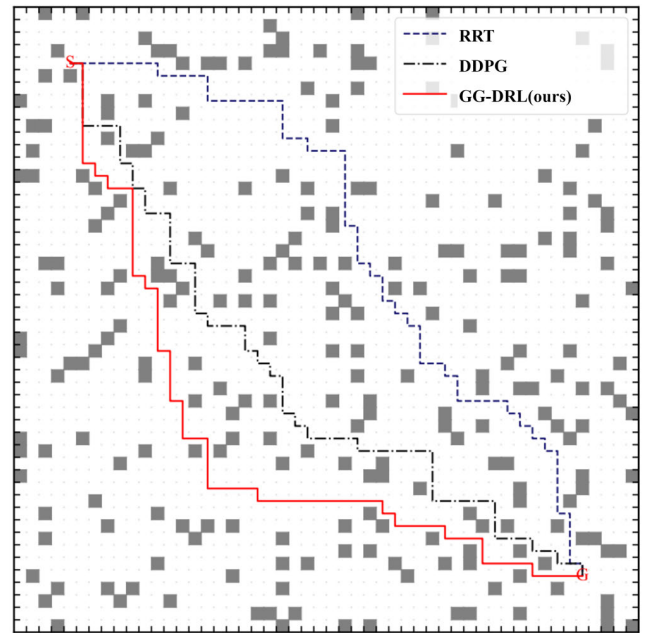


FIGURE 17. Path comparison of planning algorithms on the standard maze map.

The experiments were conducted on the standard Maze Benchmark Map. For each algorithm, 30 independent training sessions were performed, each consisting of 1,000 iterations. Four key performance metrics were evaluated: path length, computation time, obstacle avoidance times, and trajectory smoothness, to comprehensively assess the performance of the path planning algorithms.

As shown in Table 2, all three algorithms achieved the minimum path length of 81 units, indicating their capability to converge to a globally optimal path in a static environment. However, in terms of average path length, the proposed GG-DRL algorithm achieved the shortest (81.12 units), outperforming DDPG (81.37 units) and RRT (85 units), which reflects its better optimization stability and path consistency. Additionally, GG-DRL had the lowest maximum path length (86 units), compared to DDPG (87 units) and RRT (88 units), further demonstrating its trajectory reliability.

In terms of computation time, GG-DRL exhibited a significant advantage with an average of 11.94 seconds, which is substantially lower than DDPG (63.38 seconds) and RRT (297.08 seconds), representing reductions of approximately 81.17% and 95.98%, respectively. These results indicate that

TABLE 2. Comparison of path planning algorithms on the standard maze map (30 Trials).

Metric	Statistic	RRT	DDPG[44]	GG-DRL
Path Length (unit)	min	81	81	81
	max	88	87	86
	mean	85	81.37	81.12
Computation Time (s)	min	274.8	58.29	11.1
	max	320.1	70.58	12.8
	mean	297.08	63.38	11.94
Obstacle Avoidance Times	min	17	6	3
	max	30	14	9
	mean	24.3	9.8	5.2
Trajectory Smoothness	min	0.72	0.75	0.78
	max	0.84	0.88	0.91
	mean	0.79	0.82	0.86

GG-DRL achieves faster convergence and greater computational efficiency.

For obstacle avoidance performance, GG-DRL again outperformed the other algorithms, with an average of 5.2 avoidance maneuvers, significantly fewer than DDPG (9.8) and RRT (24.3). Its minimum and maximum values (3 and 9) are also notably lower than those of DDPG (6 and 14) and RRT (17 and 30), indicating that GG-DRL has a more stable and precise obstacle avoidance strategy, effectively reducing potential collisions in complex environments.

In terms of trajectory smoothness, GG-DRL achieved the highest performance, with a minimum value of 0.78 and a maximum of 0.91, compared to DDPG (0.75–0.88) and RRT (0.72–0.84). This indicates that GG-DRL not only produces feasible paths but also generates smoother and more continuous trajectories, contributing to reduced vibration, energy consumption, and mechanical wear during AGV operation.

In summary, the GG-DRL algorithm outperforms both traditional path planning methods (e.g., RRT) and classical reinforcement learning approaches (e.g., DDPG) in multiple dimensions, including path quality, computational efficiency, obstacle avoidance, and trajectory smoothness. These results validate the effectiveness and practical applicability of the proposed globally guided deep reinforcement learning method in complex static environments.

In this study, to evaluate the adaptability and robustness of path planning algorithms in dynamic environments, three test scenarios with varying levels of complexity were designed. Specifically, 10, 20, and 30 dynamic obstacles were randomly added to randomly generated static obstacle maps, forming dynamic obstacle environments as illustrated in Figure 18. Each dynamic obstacle moves back and forth along the shortest path between a designated start and goal point, which is computed using the A* algorithm, thereby simulating continuously changing interference sources in real-world environments.

Figure 18(a) depicts the scenario containing 10 dynamic obstacles. In this relatively low-complexity environment, the distribution of dynamic obstacles is sparse, resulting in minimal interference with path planning. Under such conditions, the AGV has sufficient space to perform obstacle avoidance and detouring maneuvers. Figure 18(b) presents a medium-complexity scenario with 20 dynamic obstacles. The increase in obstacle density significantly elevates the environmental complexity, requiring more frequent obstacle avoidance decisions and path adjustments during navigation. This scenario serves to evaluate each algorithm's adaptability and real-time response capability. Figure 18(c) illustrates a high-complexity scenario with 30 dynamic obstacles. In this highly dynamic and uncertain environment, the AGV faces substantial challenges in obstacle avoidance, placing greater demands on the path planning algorithm in terms of environmental perception, strategy adjustment, and trajectory smoothness.

By gradually increasing the number of dynamic obstacles across these scenarios, the performance and stability of each path planning algorithm can be comprehensively assessed under varying levels of dynamic environmental complexity.

In the test environment containing 10 dynamic obstacles, this study conducted a comparative analysis of the performance of RRT, DDPG, and the proposed GG-DRL algorithm across four key evaluation metrics: path length, computation time, obstacle avoidance times, and trajectory smoothness. The detailed results are presented in Table 3.

In terms of path length, all three algorithms successfully generated feasible paths to the goal point, with the shortest path length reaching 81 units for both DDPG and GG-DRL, and 82 units for RRT, indicating that all methods retain strong global optimization capabilities. However, in terms of average path length, GG-DRL achieved an average of 82.9 units, which is slightly better than DDPG (83.1) and RRT (82.7). This suggests that GG-DRL is more stable in generating near-optimal paths under dynamic obstacle disturbances, effectively reducing the variance in path quality.

Regarding computational efficiency, GG-DRL demonstrated a significant advantage, with an average computation time of 13.08 seconds, which is markedly lower than that of DDPG (67.19 seconds) and RRT (308.21 seconds). This highlights GG-DRL's high efficiency in solving dynamic path planning problems, primarily due to its effective global guidance mechanism and dynamic reward strategy, which substantially reduce the search space and computational complexity.

In terms of obstacle avoidance performance, GG-DRL exhibited stronger predictive and evasive capabilities when dealing with dynamic obstacles. It recorded an average of 4 avoidance maneuvers, outperforming both DDPG (4.8 times) and RRT (5.1 times). This further demonstrates GG-DRL's superior decision-making quality and responsiveness in complex dynamic environments.

Additionally, for the trajectory smoothness metric, GG-DRL achieved the highest average smoothness score of 0.88,

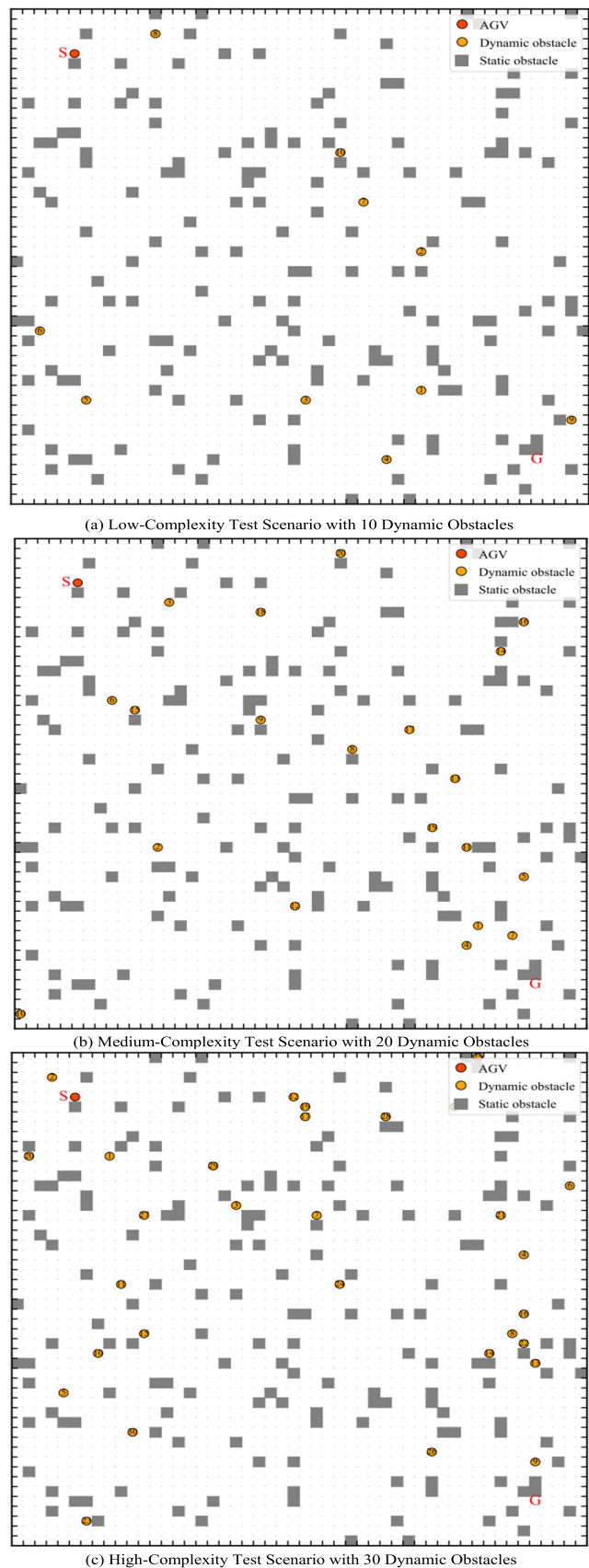


FIGURE 18. Test environments with varying dynamic obstacle densities.

TABLE 3. Statistical performance metrics in a 10 dynamic obstacles scenario.

Metric	Statistic	RRT	DDPG[44]	GG-DRL
Path Length (unit)	Min	82	81	81
	Max	84	85	85
	Mean	82.7	83.1	82.9
Computational Time(s)	Min	301.08	64.54	12.29
	Max	324.78	70.88	13.64
	Mean	308.21	67.19	13.08
Obstacle Avoidance Times	Min	3	3	2
	Max	7	6	5
	Mean	5.1	4.8	4
Trajectory Smoothness	Min	0.75	0.76	0.78
	Max	0.9	0.92	0.93
	Mean	0.85	0.86	0.88

compared to RRT (0.85) and DDPG (0.86), with a maximum smoothness value reaching 0.93. This indicates that the trajectories generated by GG-DRL offer better motion continuity and control stability, which are beneficial for reducing energy consumption and improving operational stability in real-world AGV applications.

In conclusion, within the complex path planning scenario involving 10 dynamic obstacles, the proposed GG-DRL algorithm demonstrated outstanding performance across all key metrics—path quality, computational efficiency, obstacle avoidance, and trajectory smoothness—highlighting its significant potential for efficient and reliable dynamic path planning tasks.

In the more complex testing environment with 20 dynamic obstacles, this study further compares the performance of RRT, DDPG, and the proposed GG-DRL algorithm in terms of path length, computation time, obstacle avoidance times, and trajectory smoothness. The detailed results are summarized in Table 4.

With respect to path length, all three algorithms successfully completed the navigation tasks, achieving the shortest possible path length of 81 units for both DDPG and GG-DRL, and 82 units for RRT. This indicates that each algorithm retains a strong path optimization capability even under increased environmental complexity. However, an analysis of the average path length reveals that GG-DRL achieved 83.9 units, positioned between DDPG (84.4 units) and RRT (83.4 units). This suggests that GG-DRL maintains a notable advantage in terms of path quality and optimization stability, despite the presence of dynamic disturbances.

In terms of computational efficiency, GG-DRL demonstrated exceptional performance with an average computation time of only 14.06 seconds—significantly lower than that of DDPG (76.44 seconds) and RRT (340.46 seconds). This

TABLE 4. Statistical performance metrics in a 20 dynamic obstacles scenario.

Metric	Statistic	RRT	DDPG	GG-DRL
Path Length (unit)	Min	82	81	81
	Max	89	89	88
	Mean	83.4	84.4	83.9
Computational Time (s)	Min	320.40	65.10	13.03
	Max	367.20	81.78	15.5
	Mean	340.46	76.44	14.06
Obstacle Avoidance Times	Min	5	5	4
	Max	10	9	7
	Mean	7.3	6.5	5.8
Trajectory Smoothness	Min	0.76	0.78	0.8
	Max	0.91	0.92	0.93
	Mean	0.84	0.86	0.87

notable difference highlights GG-DRL’s superior real-time responsiveness and efficiency in handling complex dynamic environments, making it more suitable for time-sensitive real-world applications.

Regarding obstacle avoidance performance, statistical analysis shows that GG-DRL required an average of only 5.8 avoidance maneuvers, compared to 6.5 for DDPG and 7.3 for RRT. This result indicates that GG-DRL incorporates a more precise and effective strategy for dynamic obstacle avoidance, significantly reducing unnecessary detours and frequent trajectory adjustments.

Additionally, in terms of trajectory smoothness, GG-DRL once again outperformed the other methods, achieving an average smoothness score of 0.87, higher than DDPG (0.86) and RRT (0.84). This demonstrates GG-DRL’s advantage in maintaining trajectory continuity and stability, which is beneficial for reducing vibrations, lowering energy consumption, and improving overall transportation safety and reliability in actual AGV operations.

In summary, within the complex navigation scenario involving 20 dynamic obstacles, the proposed GG-DRL algorithm exhibited superior performance across all key metrics—path optimization, computational efficiency, dynamic obstacle avoidance, and trajectory smoothness—demonstrating enhanced robustness and broader practical application potential.

In the highly complex scenario with 30 dynamic obstacles, this study conducted a detailed performance analysis of RRT, DDPG, and the proposed GG-DRL algorithm across four key metrics: path length, computation time, obstacle avoidance times, and trajectory smoothness. The specific results are presented in Table 5.

With respect to path length, all three algorithms successfully completed the navigation task, achieving the shortest path length of 81 units, indicating that their fun-

damental path-searching capabilities were not significantly impaired by the increased environmental complexity. However, in terms of average path length, GG-DRL recorded a value of 84.4 units, which is slightly higher than DDPG (81.9 units) and RRT (81.6 units). This discrepancy is primarily attributed to GG-DRL’s strategy of prioritizing AGV operational safety and path stability by moderately sacrificing path optimality in favor of more efficient dynamic obstacle avoidance in dense environments.

In terms of computation time, GG-DRL demonstrated a clear advantage with an average time of 15.09 seconds, which is substantially lower than DDPG (71.57 seconds) and RRT (327.01 seconds). These results indicate that GG-DRL maintains excellent real-time responsiveness and high computational efficiency, even under highly dynamic and complex conditions, showcasing strong adaptability and practical application potential.

Regarding obstacle avoidance performance, GG-DRL achieved an average of 8.8 avoidance maneuvers, significantly fewer than DDPG (10.9 times) and RRT (12.2 times). This highlights GG-DRL’s superior dynamic obstacle prediction accuracy and robust path adjustment strategy, which effectively reduces the potential risk of collisions between the AGV and moving obstacles, thereby enhancing operational safety.

In terms of trajectory smoothness, GG-DRL once again outperformed the other two algorithms, with an average smoothness score of 0.87, compared to 0.85 for DDPG and 0.83 for RRT. This indicates that GG-DRL not only effectively accomplishes path planning tasks but also produces smoother and more continuous trajectories, which help reduce energy consumption and mechanical wear during AGV operation.

In conclusion, even in an extremely complex environment with 30 dynamic obstacles, the GG-DRL algorithm continues to deliver outstanding overall performance in terms of path optimization, computational efficiency, dynamic obstacle avoidance, and trajectory smoothness. These results further validate its superiority and practical applicability for reliable path planning in highly dynamic and uncertain scenarios.

III. RESULTS AND DISCUSSION

This chapter aims to introduce how to build a simulation environment based on ROS+Gazebo and set key experimental parameters to verify the feasibility and effectiveness of the multi-AGV dynamic path planning algorithm under global guidance based on deep reinforcement learning. In this process, apart from the main AGV, the other AGVs are considered as dynamic obstacles to further explore the comparison between the proposed algorithm and the traditional artificial potential field method in terms of dynamic obstacle avoidance capability. Moreover, this study also aims to test the obstacle avoidance performance of the AGV after successfully avoiding obstacles, facing newly emerged static obstacles, i.e., its adaptability to environmental changes.

TABLE 5. Statistical performance metrics in a 30 dynamic obstacles scenario.

Metric	Statistic	RRT	DDPG	GG-DRL
Path Length (unit)	Min	81	81	81
	Max	83	83	88
	Mean	81.6	81.9	84.4
Computational Time(s)	Min	315.51	64.54	13.51
	Max	341.07	73.68	16.09
	Mean	327.01	71.57	15.09
Obstacle Avoidance Times	Min	8	7	6
	Max	15	13	11
	Mean	12.2	10.9	8.8
Trajectory Smoothness	Min	0.72	0.74	0.76
	Max	0.9	0.92	0.94
	Mean	0.83	0.85	0.87

A. SET UP A SIMULATION ENVIRONMENT

The experimental environment constructed in this study runs on the Linux operating system, using ROS as the middleware. Specifically, the system used for the experiments is Ubuntu 16.04, while the deep learning environment is configured with Cuda and Cudnn 7.1.4, along with Tensorflow-gpu version 1.12.0. The version of ROS used is ROS Kinetic. It is worth noting that although this study used ROS 1 Kinetic and Ubuntu 16.04 platforms, the proposed method is theoretically platform independent. In future work, we will strive to migrate the system to ROS 2 and Ubuntu 24.04 to verify the impact of the new platform on overall performance and explore further algorithm optimization using the advantages of the new platform. In terms of hardware configuration, the experimental environment is equipped with an NVIDIA GeForce GTX 1080 Ti graphics card, featuring 11GB of video memory, paired with an Intel Xeon E5-2680 v3 processor, 32GB of DDR4 memory, and a 1TB NVMe solid-state drive. The simulation environment built on ROS and Gazebo [41] is detailed in Figure 19.

Figure 14 showcases the interface of the Gazebo simulation environment, where the left side presents the simulation interface itself. Through the ratio of simulation time to real time combined with the preset AGV speed, the actual working distance of the AGV is calculated. In this simulation system, there exists a main AGV, and the right side of the figure displays the real-time images captured by the environmental perception module of the main AGV during the path planning process. In this environment, besides the main AGV, there are other AGVs engaged in cargo transportation tasks.

In a rail transport environment, the simultaneous transportation tasks of multiple AGVs require these robots to be able to avoid obstacles in real-time, which poses certain demands on the flexibility of the AGVs. Considering various situations AGVs might face in real operations, this study has designed multiple executable actions for AGVs. Specifically,

TABLE 6. AGV execution.

The action number	Velocity (m/s)	angular velocity (rad/s)	Actual speed response(s)	Actual measured velocity(m/s)
0	0	0	0.00	0.00
1	0	1	0.00	0.00
2	0	-1	0.00	0.00
3	1	0	0.50	0.98
4	1	0.5	0.52	0.95
5	1	-1	0.48	1.00
6	1	-0.5	0.50	0.97
7	2	0	0.30	1.97
8	2	0.3	0.32	1.95
9	2	-0.3	0.31	1.98

when the AGV speed is low, it can more easily turn when encountering obstacles; in a hovering state (speed of 0 m/s), the AGV is set with a larger angular velocity to quickly adjust direction; and at high speed (speed of 2 m/s), it is set with a smaller angular velocity to avoid accidents during emergency obstacle avoidance. The specific settings of these actions are detailed in Table 6.

Moreover, we recorded the actual velocity response of the AGVs during each action and included this information as the new column “Actual Velocity Response” in Table 1. For instance, when performing action 3, the AGV accelerates from a stationary state to the target speed of 1 m/s in 0.5 seconds; for action 7, it reaches the target speed of 2 m/s in only 0.3 seconds. These data visually demonstrate how different action parameters influence the dynamic response of the AGV. Additionally, the column “Experimentally Measured Speed” was introduced to reflect the actual speeds achieved by the AGVs during the stable operating phase. Experimental results indicate that the actual speeds closely match the preset target speeds, with deviations maintained within an acceptable range. For example, in action 3, the set target speed was 1 m/s, whereas the measured speed was approximately 0.98 m/s; similarly, action 7 had a target speed of 2 m/s, with an experimentally measured speed of about 1.97 m/s. These outcomes confirm that the proposed control algorithm effectively satisfies the design requirements during both acceleration and stable operating stages, validating the rationality of the action parameter settings. With these experimental insights, system parameters can be precisely fine-tuned, further enhancing the AGV’s dynamic response performance and overall operational stability.

B. EXPERIMENTAL SETUP

1) ALGORITHM PARAMETER SETTINGS

During the AGV algorithm validation phase, the setting of parameters is crucial to ensure that the model can effectively learn and converge. As described in Table 7, we set the maximum training steps for the AGV at 50,000 steps. This setting aims to prevent the model from failing to achieve

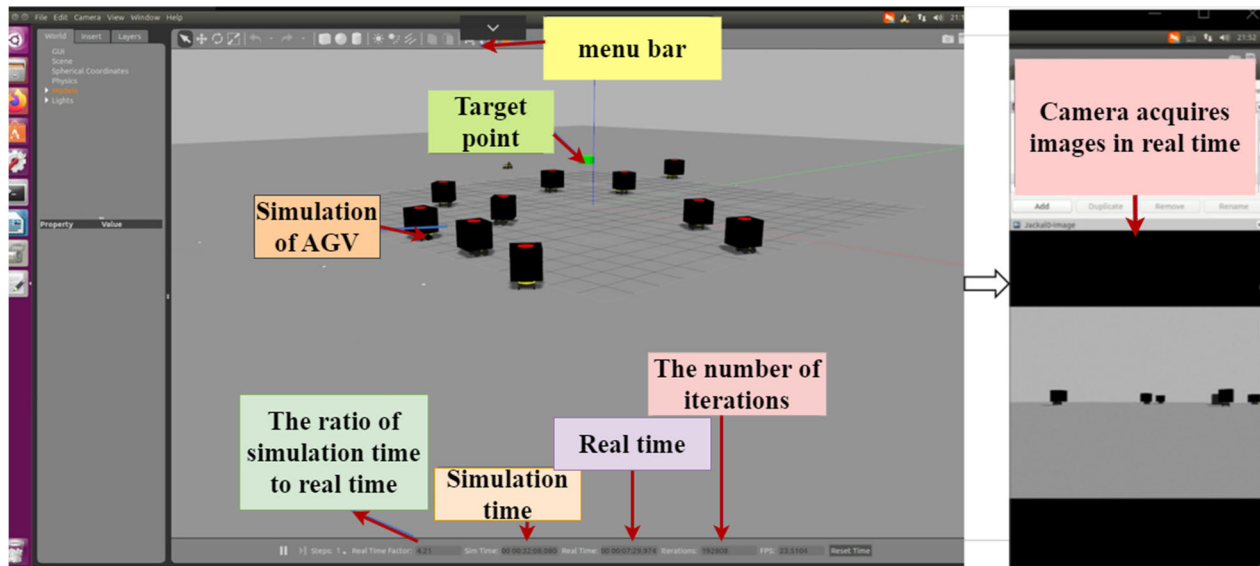


FIGURE 19. Simulation environment.

TABLE 7. Parameters of the verification algorithm are verified.

Parameter	Size
Learning rate Learn_rate	0.00001
Attenuation factor γ	0.95
Max_episodes	700
Target_updatenum	700
mini-batch	64
Max_steps	50000
Action_num	8
Greedy policy termination parameter ϵ_{final}	0.1
Rrplaybuffer_num	7000

the desired learning effect due to insufficient training steps. Additionally, the capacity of the experience replay pool is also set to a large value, with the purpose of maximizing the use of historical training data. This method helps to reduce the correlation between data samples, ensuring the model can learn from a variety of different scenarios, thereby improving the efficiency and effectiveness of the training process.

2) REWARD FUNCTION PARAMETER SETTINGS

The design of the AGV's reward function is a key task in reinforcement learning, as it directly influences the direction and efficiency of the model's learning. The size of the reward is determined based on the distance of the AGV to the target point and the relative deviation angle to the target point. Specifically, when the distance between the AGV and the target point is less than 4 meters, the set reward factor is larger than when the distance is greater than 4 meters. Such a design motivates the AGV to accelerate its movement speed when approaching the target point, to reach the target point more quickly.

During the training process of the AGV, if the AGV collides with other AGVs, a significant penalty is given, and the current training set is terminated, to reinforce the behavior strategy of avoiding collisions. Conversely, when the AGV successfully reaches the target point, it receives a substantial positive reward, and the result is stored in the experience pool. This approach not only rewards the correct behavior but also, through the mechanism of experience replay, allows the model to adopt the same strategy when encountering similar scenarios later, thus accelerating the model's convergence process.

Similar to the reward setting for the distance to the target point, the reward setting for the AGV's relative deviation angle to the target point follows similar principles, i.e., by adjusting reward parameters, motivating the AGV to reduce its deviation angle to the target point, thus arriving at the target location more accurately. The specific numerical settings of the reward parameters are detailed in Table 8. The design of this reward function takes into account the practical needs of AGV operations, promoting more efficient and precise behavior of the AGV by incentivizing two key factors: distance and angle.

C. DISPLAY OF MULTI-AGV PATH PLANNING RESULTS IN DYNAMIC ENVIRONMENT

1) CHANGES IN THE REWARD FUNCTION

Through observation and analysis of the training process of AGVs at different stages of deep reinforcement learning, we can obtain a series of valuable insights, especially regarding methods to improve model performance and learning efficiency. During the initial training process, although the AGV achieved high reward values in several episodes, indicating its success in reaching the target point in these cases, the model as a whole did not show signs of convergence (as shown in Figure 20-a). This might mean that, although the

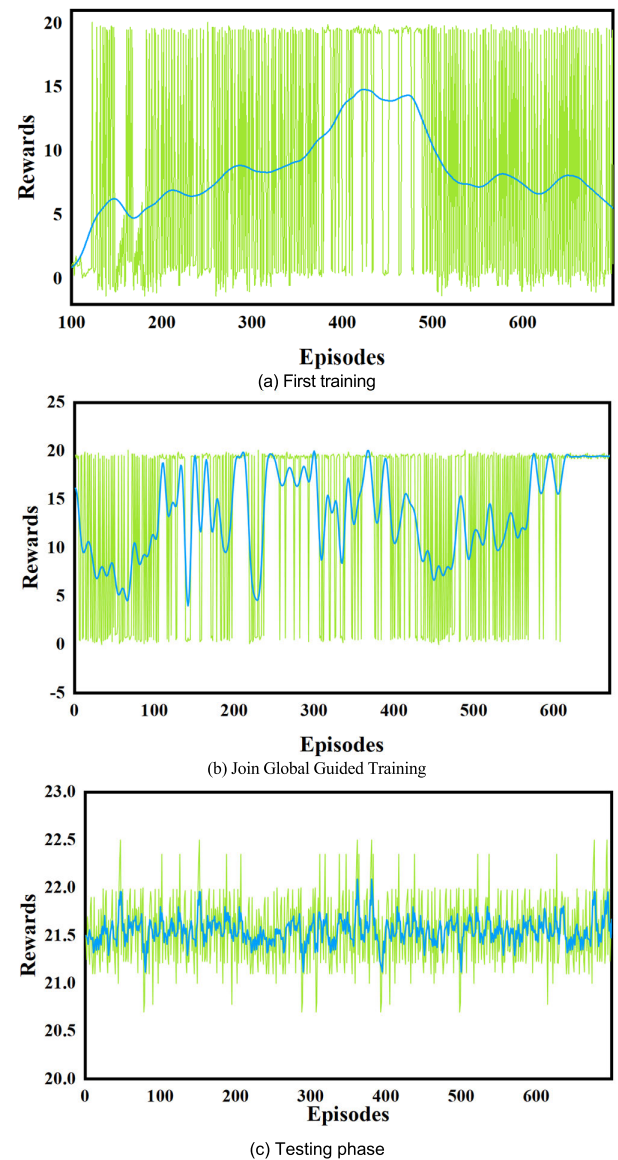
TABLE 8. Basic settings of AGV reward functions.

The scene in which the AGV is located	Reward function parameter settings
More than 4 meters from the target point	$0.2 \times (d_{g_{t+1}} - d_{g_t})$
Within 4 meters of the target point	$0.5 \times (d_{g_{t+1}} - d_{g_t})$
Within 0.5m of the obstacle	$-0.1 \times (d_{o_{t+1}} - d_{o_t})$
Collision with obstacles	-5, End that episodes
Reach within 0.5m of the target point	20, End that episodes
The deviation angle between the AGV and the target point is greater than 60°	$-0.3 \times (\theta_{g_{t+1}} - \theta_{g_t})$
The deviation angle between the AGV and the target point is within 20°	$0.1 \times (\theta_{g_{t+1}} - \theta_{g_t})$

AGV learned to navigate successfully in specific situations, it failed to generalize these strategies to a wider range of environments or scenarios, resulting in its overall performance still being unstable. In the second training session, the introduction of a global guidance strategy significantly accelerated the learning process of the model, ultimately achieving convergence (as shown in Figure 20-b). Global guidance likely provided additional contextual information or more effective learning strategies, helping the model more quickly identify and learn the optimal paths to the target point, thus improving training efficiency and the stability of the model. In actual testing processes, the trend graph of reward values over episodes (as shown in Figure 20-c) demonstrates that the reward values gradually converge in dynamic stability, indicating that the AGV successfully achieves goal navigation through continuous learning and adaptation. This stable trend indicates that the AGV not only learned how to successfully reach the target point but also, through continuous learning and adjustments, its strategies have become stable enough to adapt to various dynamically changing environments. Overall, these observations highlight the important role of global guidance in accelerating model training and enhancing model generalization capabilities. They also show that through continuous learning and optimization, deep reinforcement learning models can gradually improve their performance, achieving stable and effective path planning and goal navigation in complex and dynamic environments.

2) TRAJECTORY RESULTS OF MULTI-AGV DYNAMIC PATH PLANNING

By setting up experiments with 10 Automated Guided Vehicles (AGVs) for dynamic path planning, where the starting and target points of each AGV are randomly generated, we can deeply explore and validate the efficiency and effectiveness of the proposed dynamic path planning algorithm. From the presentation in Figure 21(a)-(h), we selected displays of 8 path planning results, with each color representing the path trajectory of an AGV, and numbers indicating time steps. This visual representation allows us to intuitively observe that, even in different map environments and with multiple AGVs performing dynamic path planning simulta-

**FIGURE 20.** Graph of the value of the reward function in the dynamic obstacle environment with episode.

neously, no conflicts occurred. This result emphasizes the capability of the proposed algorithm in handling dynamic obstacle avoidance and path planning in multi-AGV systems.

Furthermore, Figure 22(a)-(d) presents the path planning results from the perspective of the main AGV, with other AGVs acting as dynamic obstacles. In this perspective, different colors represent different dynamic AGVs, while the trajectory of the main AGV is shown in red, with numbers also representing different moments. Through this analysis perspective, we can see that even in complex scenarios where other AGVs are considered dynamic obstacles, the main AGV could effectively avoid these obstacles without any collisions. The analytical results from these two perspectives not only validate the effectiveness of the proposed dynamic path planning algorithm but also demonstrate the algorithm's potential in handling complex dynamic environments in practical applications. These findings provide valuable references and insights for future applications and research in similar multi-AGV systems.

D. MULTI-AGV PROFICIENCY VERIFICATION IN THE SIMULATION ENVIRONMENT

1) DYNAMIC OBSTACLE AVOIDANCE ABILITY VERIFICATION

This experiment conducted in the simulation environment aims to verify the obstacle avoidance capabilities of a main AGV in an environment coexisting with 10 dynamic AGVs performing transportation tasks. Through a specific simulation scenario (as shown in Figure 23), we can deeply analyze the behavior and decision-making process of the main AGV. At $t=10$ seconds, the main AGV detects a dynamic AGV (AGV1) moving ahead through its perception system. Subsequently, at $t=11$ seconds, the main AGV further perceives that continuing on its current path would result in a collision with AGV1. To avoid this collision, the main AGV decides to execute an avoidance action, i.e., passing to the left of AGV1. This decision and action demonstrate the main AGV's rapid response and effective obstacle avoidance capability towards dynamic obstacles.

By $t=20$ seconds, the main AGV successfully reaches a position only 0.5 meters away from the target point, thereby receiving a reward and marking the end of the current episode. At this moment, the red arrow points to the image captured by the camera as the main AGV is about to reach the target point, which not only provides intuitive proof of successful obstacle avoidance but also shows the visual perception capability of the main AGV when approaching the target point. This simulation experiment not only verifies the main AGV's obstacle avoidance capability in complex dynamic environments but also reflects the application potential of deep reinforcement learning algorithms in real-time decision-making and path planning. Through this method, AGVs can complete tasks efficiently while ensuring safety.

2) AUTONOMOUS PROFICIENCY TESTING

In this more complex simulation experiment setup, in addition to the 10 dynamic AGVs carrying goods and 1 main AGV, new static obstacles were introduced (as shown in Figure 24) to test the AGV's obstacle avoidance capabilities

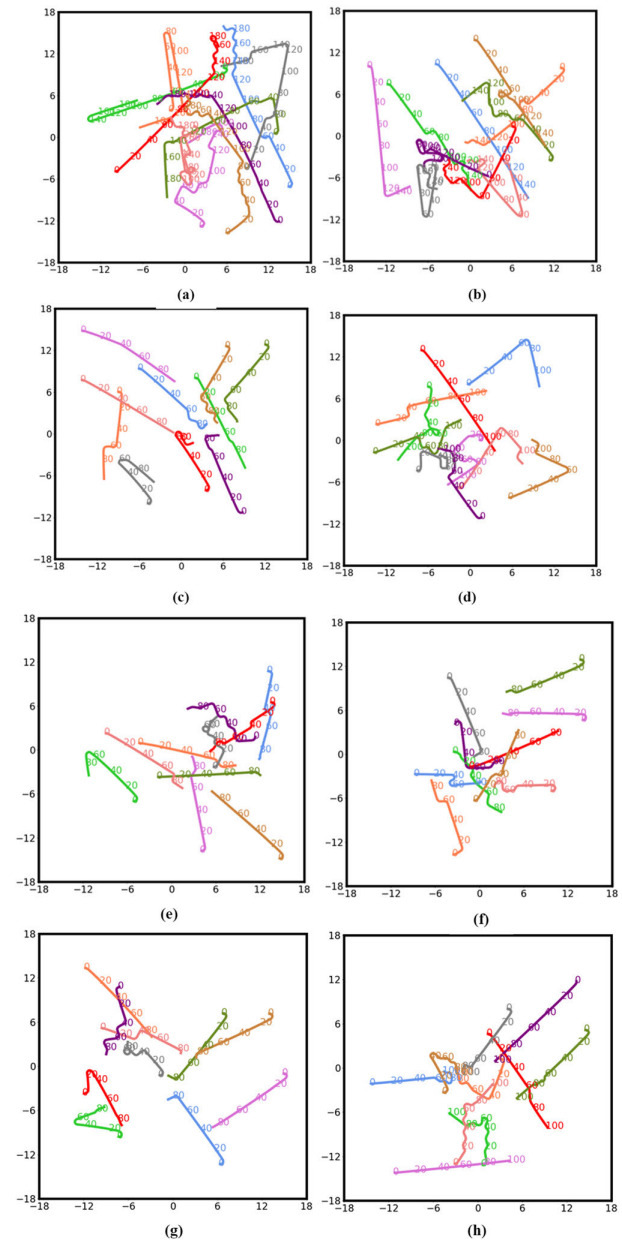


FIGURE 21. Trajectory diagram of dynamic path planning for multiple AGVs.

when encountering sudden static obstacles. The grey cubes represent the newly introduced static obstacles, and the right side of Figure 24 displays the real-time visual information captured by the main AGV's camera, reflecting how the AGV perceives obstacles in front of it.

After recognizing the obstacle, the main AGV needs to make a quick decision to avoid a potential collision. According to Figure 25, the AGV adopted a steering strategy and began to circumnavigate the obstacle. As the AGV moved, the grey obstacle gradually faded out of the AGV's field of view until, at the time step shown in Figure 26, the static obstacle was completely out of the AGV's view, indicating that the AGV had successfully avoided the obstacle and continued towards the target point.

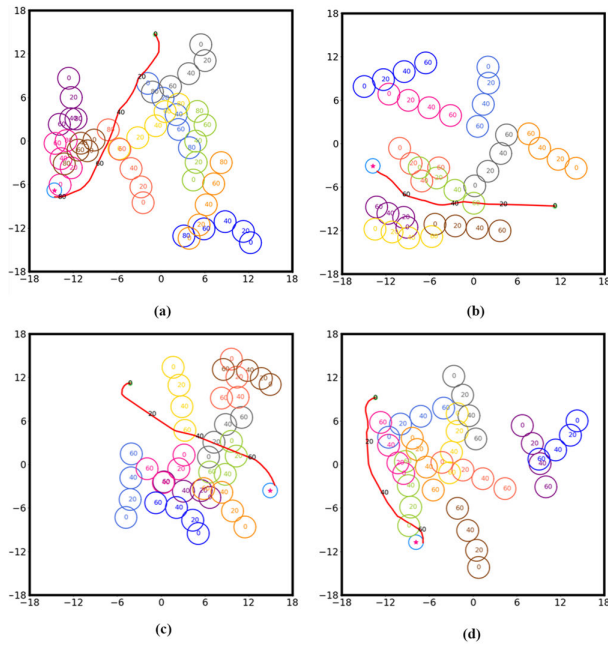


FIGURE 22. Dynamic obstacle trajectory diagram.

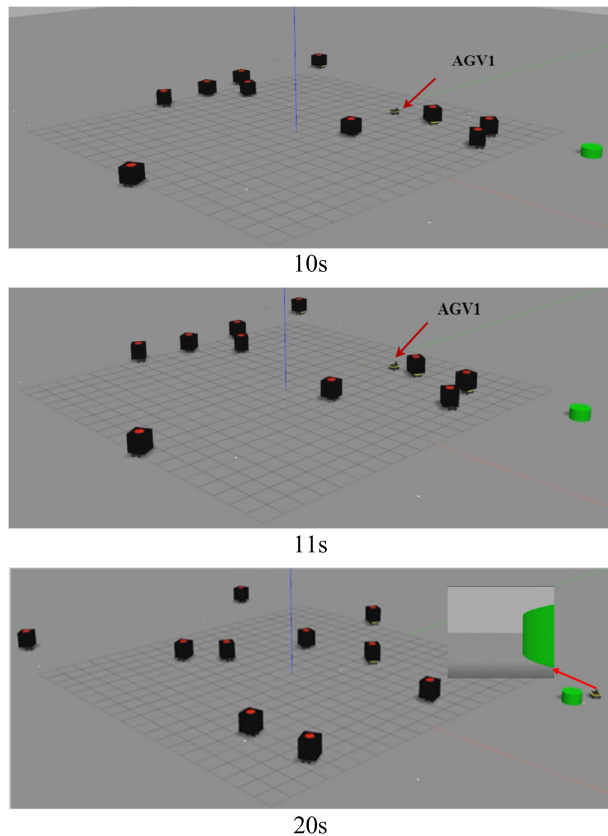


FIGURE 23. Obstacle avoidance capability test.

This series of actions and decision-making processes not only demonstrates the main AGV's rapid adaptability to changes in the environment but also showcases its efficient obstacle avoidance strategy. Through real-time processing of visual information and decision-making, the AGV can effectively deal with static obstacles that appear on the path,

ensuring its safety and smoothness during task execution. This capability is crucial for the application of multi-AGV systems in complex and unpredictable environments, especially in logistics and manufacturing scenarios that require high automation and flexibility.

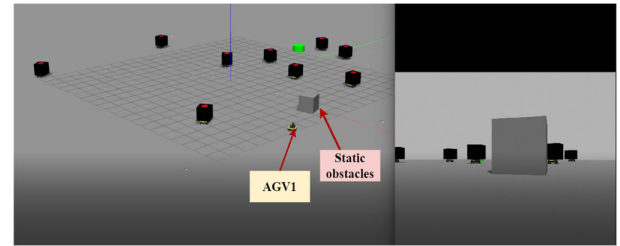


FIGURE 24. Introducing static obstacles in a dynamic environment.

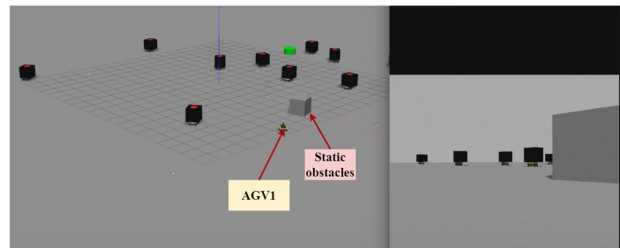


FIGURE 25. AGV performs steering action.

In this section, through meticulously designed simulation experiments, the path planning and obstacle avoidance capabilities of multi-Automated Guided Vehicle (AGV) systems in dynamic environments were comprehensively validated. These experiments not only demonstrated how multiple AGVs can perform effective global planning and trajectory optimization in dynamic environments but also particularly emphasized the AGVs' autonomous obstacle avoidance capabilities in the face of sudden static obstacles. The experimental results clearly showed that the proposed algorithm could achieve collision-free path planning for multiple AGVs in dynamic environments, fully meeting the material handling requirements in rail transport environments. In such rail transport environments, AGVs need to avoid dynamic obstacles (such as other AGVs) and must also be able to effectively avoid sudden static obstacles, ensuring the smoothness and safety of the entire transport process. To meet this challenge, this study conducted extensive validations under a multi-AGV simulation platform built for rail transport material transportation. The validation results once again confirmed that even in existing complex rail transport environments, the AGV system could successfully identify and avoid newly emerged static obstacles, demonstrating its outstanding environmental adaptability and obstacle avoidance efficiency. Through these experiments, the algorithm of this paper has proven its efficiency and reliability in actual application scenarios, especially in modern logistics and manufacturing industries that require high automation and flexible responses to environmental changes. These

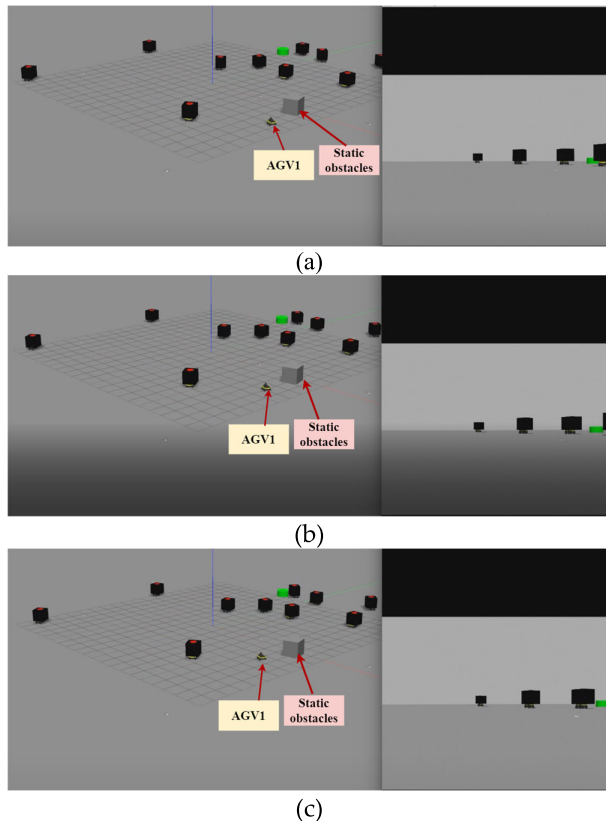


FIGURE 26. AGV avoids static obstacles.

achievements not only provide valuable experience for further research and development of AGV systems but also lay a solid foundation for technological innovation and application in related fields.

IV. CONCLUSION

This study is focused on the specific context of material handling within rail transportation environments, investigating the limitations presented by current path planning algorithms in this setting. By incorporating a multi-autonomous guided vehicle (AGV) path planning approach based on visual understanding, this research aims to enhance the adaptability and efficiency of path planning. Specifically, the study treats the fixed tracks within the rail system as static obstacles and has developed a global path planning strategy for multiple AGVs. Utilizing data collected from AGVs' onboard cameras and LiDAR sensors, a multi-sensor data fusion-based environmental perception module was developed. Furthermore, leveraging the strengths of Deep Learning (DL) and Reinforcement Learning (RL), a deep reinforcement learning framework was proposed, where DL facilitates environmental perception and RL oversees decision-making. This framework enables collision-free path planning by rewarding or penalizing the AGVs based on the outcome of their actions in the environment. Experimental results demonstrate the capability of this research to meet the complex material handling requirements in rail transportation environments,

providing AGVs with autonomous obstacle avoidance and navigation abilities, and showcasing significant application potential. Additionally, the AGVs exhibited high reliability and safety during the handling process, promising a substantial reduction in manual handling costs, hence possessing significant economic value. The main contributions of this research include: 1. An end-to-end waypoint fitting heuristic algorithm, based on the A* algorithm, tailored for the uniqueness of rail transportation environments. This algorithm integrates conflict resolution mechanisms and trajectory optimization capabilities, planning global paths for each AGV. 2. A multi-sensor feature fusion environmental perception algorithm developed based on data from AGVs' onboard cameras and LiDAR sensors. This algorithm effectively perceives and identifies dynamic carts, static obstacles, and target points within rail transportation environments. 3. A reinforcement learning framework built upon Deep Q-Networks (DQN), introducing global guidance in the training phase to accelerate model convergence. Based on this foundational DRL model, novel reward functions were designed, and, in conjunction with the aforementioned environmental perception module, dynamic obstacle avoidance and multi-AGV path planning were achieved. 4. To further investigate the path planning capabilities of AGVs in complex environments, a highly simulated testing environment was constructed. Through this simulation platform, the obstacle avoidance performance of the multi-AGV system under simulated conditions was meticulously verified, especially its adaptive mechanism in response to newly emerging obstacles. This research not only demonstrates AGVs' ability to handle sudden obstacles in simulated rail transport environments but also provides strong experimental support for efficient material handling needs in the context of rail transportation.

REFERENCES

- [1] L. Jia, P. Cheng, Z. Zhang, L. Ji, and C. Xu, "Integrated development of rail transit and energies in China: Development paths and strategies," *Chin. J. Eng. Sci.*, vol. 24, no. 3, p. 173, 2022.
- [2] M. Yang, Y. Bian, L. Ma, G. Liu, and H. Zhang, "Research on traffic control algorithm based on multi-AGV path planning," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2021, pp. 697–702.
- [3] A. Y. Pratama, M. R. Ariyadi, M. N. Tamara, D. S. Purnomo, N. A. Ramadhan, and B. Pramujati, "Design of path planning system for multi-agent AGV using A* algorithm," in *Proc. Int. Electron. Symp. (IES)*, Aug. 2023, pp. 335–341.
- [4] J. Hu, W. Shang, and H. Lou, "Research on AGV path based on optimal planning," in *Proc. 4th Int. Conf. Data Sci. Inf. Technol.*, Jul. 2021, pp. 236–240.
- [5] T. N. Tien and K.-V. Nguyen, "Updated weight graph for dynamic path planning of multi-AGVs in healthcare environments," in *Proc. Int. Conf. Adv. Technol. Commun. (ATC)*, Oct. 2022, pp. 130–135.
- [6] F. Huang, W. Guo, and H. Zhao, "AGV path planning based on improved genetic algorithm," in *Proc. 2nd Int. Symp. Control Eng. Robot. (ISCER)*, Feb. 2023, pp. 3323–3327.
- [7] J. Shi, W. Wang, X. Wang, H. Sun, X. Lan, J. Xin, and N. Zheng, "Leveraging spatio-temporal evidence and independent vision channel to improve multi-sensor fusion for vehicle environmental perception," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 591–596.
- [8] N. Senel, K. Kefferpütz, K. Doycheva, and G. Elger, "Multi-sensor data fusion for real-time multi-object tracking," *Processes*, vol. 11, no. 2, p. 501, Feb. 2023.

- [9] M. Bouain, K. M. A. Ali, D. Berdjag, N. Fakhfakh, and R. B. Atitallah, "An embedded multi-sensor data fusion design for vehicle perception tasks," *J. Commun.*, vol. 13, no. 1, pp. 8–14, 2018.
- [10] C. Xiang, C. Feng, X. Xie, B. Shi, H. Lu, Y. Lv, M. Yang, and Z. Niu, "Multi-sensor fusion and cooperative perception for autonomous driving: A review," *IEEE Intell. Transp. Syst. Mag.*, vol. 15, no. 5, pp. 36–58, Sep. 2023.
- [11] M. Jayaratne, D. de Silva, and D. Alahakoon, "Unsupervised machine learning based scalable fusion for active perception," *IEEE Trans. Autom. Sci. Eng.*, vol. 16, no. 4, pp. 1653–1663, Oct. 2019.
- [12] X. Liao, Y. Wang, Y. Xuan, and D. Wu, "AGV path planning model based on reinforcement learning," in *Proc. Chin. Autom. Congr. (CAC)*, Nov. 2020, pp. 6722–6726.
- [13] H. Yin, Y. Lin, J. Yan, Q. Meng, K. Festl, L. Schichler, and D. Watenig, "AGV path planning using curiosity-driven deep reinforcement learning," in *Proc. IEEE 19th Int. Conf. Autom. Sci. Eng. (CASE)*, Aug. 2023, pp. 1–6.
- [14] X. Guo, Z. Ren, Z. Wu, J. Lai, D. Zeng, and S. Xie, "A deep reinforcement learning based approach for AGVs path planning," in *Proc. Chin. Autom. Congr. (CAC)*, Nov. 2020, pp. 6833–6838.
- [15] X. Ye, Z. Deng, Y. Shi, and W. Shen, "Toward energy-efficient routing of multiple AGVs with multi-agent reinforcement learning," *Sensors*, vol. 23, no. 12, p. 5615, Jun. 2023.
- [16] L. Zhang, G. Yin, J. Li, and J. Jiang, "Research on AGV path planning based on reinforcement learning," in *Proc. 8th Int. Conf. Intell. Comput. Signal Process. (ICSP)*, Apr. 2023, pp. 934–937.
- [17] P. Theodorou, K. Tsiligkos, and A. Meliones, "Multi-sensor data fusion solutions for blind and visually impaired: Research and commercial navigation applications for indoor and outdoor spaces," *Sensors*, vol. 23, no. 12, p. 5411, Jun. 2023.
- [18] X. Cao, B. Tan, Y. Li, and S. Ding, "Dynamic load regulation of robots with multi-sensor fusion," *J. Phys., Conf. Ser.*, vol. 2400, no. 1, Dec. 2022, Art. no. 012022.
- [19] D. J. Yeong, G. Velasco-Hernandez, J. Barry, and J. Walsh, "Sensor and sensor fusion technology in autonomous vehicles: A review," *Sensors*, vol. 21, no. 6, p. 2140, Mar. 2021.
- [20] Y. Wang and H. Fang, "Estimation of LAI with the LiDAR technology: A review," *Remote Sens.*, vol. 12, no. 20, p. 3457, Oct. 2020.
- [21] S. Zhang, Z. Chen, Y. Gao, W. Wan, J. Shan, H. Xue, F. Sun, Y. Yang, and B. Fang, "Hardware technology of vision-based tactile sensor: A review," *IEEE Sensors J.*, vol. 22, no. 22, pp. 21410–21427, Nov. 2022.
- [22] A. Chatzisavvas, M. Louta, and M. Dasygenis, "Enhancing the A* algorithm for efficient route planning of agriculture," in *Proc. 18th Int. Workshop Cellular Nanosc. Netw. Their Appl. (CNNA)*, Sep. 2023, pp. 1–4.
- [23] G. Xu, Z. Tian, and S. Cai, "An improved A* algorithm based on bidirectional search," in *Proc. Int. Conf. Mach. Learn., Cloud Comput. Intell. Mining*. Singapore: Springer, 2024, pp. 314–322.
- [24] Z. Ruifang, J. Tianyi, and Z. Haitao, "Application of improved Dijkstra algorithm in two-dimensional path planning problem," in *Proc. 4th Int. Conf. Intell. Inf. Process.*, Nov. 2019, pp. 211–215.
- [25] L. Liu, X. Wang, X. Yang, H. Liu, J. Li, and P. Wang, "Path planning techniques for mobile robots: Review and prospect," *Expert Syst. Appl.*, vol. 227, Oct. 2023, Art. no. 120254.
- [26] Y. Ke, "Comparative analysis of path planning algorithms and prospects for practical application," *Highlights Sci., Eng. Technol.*, vol. 52, pp. 202–207, Jul. 2023.
- [27] Y. Li, S. Wang, L. Jiang, Y. Xie, J. Meng, and H. Wu, "Path planning of composite trackless AGV considering map preprocessing," in *Proc. IEEE REGION Conf. (TENCON)*, Nov. 2020, pp. 194–199.
- [28] J. Yu, G. Liu, Z. Zhao, X. Wang, J. Xu, and Y. Bai, "Improved D* Lite algorithm path planning in complex environment," in *Proc. Chin. Autom. Congr. (CAC)*, Nov. 2020, pp. 2226–2230.
- [29] Y. Li, F. Yang, X. Zhang, D. Yu, and X. Yang, "Improved D* lite algorithm for ship route planning," *J. Mar. Sci. Eng.*, vol. 12, no. 9, p. 1554, Sep. 2024.
- [30] B. Yan, T. Chen, X. Zhu, Y. Yue, B. Xu, and K. Shi, "A comprehensive survey and analysis on path planning algorithms and heuristic functions," in *Proc. Intell. Comput., Comput. Conf.*, Jan. 2020, pp. 581–598.
- [31] J. Lu and Y. Sun, "Application of heuristic function optimization strategy of A* algorithm in path planning of mobile robot," in *Proc. 5th Int. Conf. Comput. Eng. Appl. (ICCEA)*, Apr. 2024, pp. 77–80.
- [32] T. Zhao, M. Kadoch, and R. Peng, "Reinforcement learning for path planning in communication-limited environments," in *Proc. IEEE Int. Symp. Broadband Multimedia Syst. Broadcast. (BMSB)*, Jun. 2024, pp. 1–5.
- [33] C. Wang and J. Mao, "Summary of AGV path planning," in *Proc. 3rd Int. Conf. Electron. Inf. Technol. Comput. Eng. (EITCE)*, Oct. 2019, pp. 332–335.
- [34] X. Yin, P. Cai, K. Zhao, Y. Zhang, Q. Zhou, and D. Yao, "Dynamic path planning of AGV based on kinematical constraint A* algorithm and following DWA fusion algorithms," *Sensors*, vol. 23, no. 8, p. 4102, Apr. 2023.
- [35] M. De Ryck, D. Pissoot, T. Holvoet, and E. Demeester, "Decentral task allocation for industrial AGV-systems with resource constraints," *J. Manuf. Syst.*, vol. 59, pp. 310–319, Apr. 2021.
- [36] A. Liu, Y. Yang, Q. Sun, and Q. Xu, "A deep fully convolution neural network for semantic segmentation based on adaptive feature fusion," in *Proc. 5th Int. Conf. Inf. Sci. Control Eng. (ICISCE)*, Jul. 2018, pp. 16–20.
- [37] S. Li, J. Yan, and L. Li, "Automated guided vehicle: The direction of intelligent logistics," in *Proc. IEEE Int. Conf. Service Oper. Logistics, Informat. (SOLI)*, Jul. 2018, pp. 250–255.
- [38] C. Zheyi and X. Bing, "AGV path planning based on improved artificial potential field method," in *Proc. IEEE Int. Conf. Power Electron., Comput. Appl. (ICPECA)*, Jan. 2021, pp. 32–37.
- [39] K. R. Jayasree, P. R. Jayasree, and A. Vivek, "Smoothed RRT techniques for trajectory planning," in *Proc. Int. Conf. Technological Advancements Power Energy (TAP Energy)*, Dec. 2017, pp. 1–8.
- [40] Y. Huang, "Deep Q-networks," in *Deep Reinforcement Learning: Fundamentals, Research and Applications*. Singapore: Springer, 2020, pp. 135–160.
- [41] M. Aksu, J. L. Michaloski, and F. M. Proctor, "Virtual experimental investigation for industrial robotics in GAZEBO environment," presented at the ASME Int. Mech. Eng. Congr. Expo. (IMECE), vol. 5, Nov. 2018, Art. no. V002T02A071.
- [42] X. Chen, S. Liu, J. Zhao, H. Wu, J. Xian, and J. Montewka, "Autonomous port management based AGV path planning and optimization via an ensemble reinforcement learning framework," *Ocean Coastal Manage.*, vol. 251, May 2024, Art. no. 107087.



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